tropical: LIS

2026-10-25T23:07:27

1 Key results

This report covers 240 runs (5 seeds; 0 diverged) under frozen hash `69813b292d06`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the parametric path: values away from the metric bounds. Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.4729	[0.464, 0.482]	0.0112	5	0
adaptive	pointer	0.01	0.4748	[0.468, 0.481]	0.0086	5	0
adaptive	pointer	0.1	0.4736	[0.466, 0.481]	0.00984	5	0
adaptive	pointer	1	0.4749	[0.468, 0.482]	0.0088	5	0
adaptive	pointer	10	0.4741	[0.467, 0.481]	0.00945	5	0
adaptive	pointer	100	0.4748	[0.468, 0.482]	0.00893	5	0
adaptive	pointer	1000	0.4754	[0.468, 0.483]	0.00916	5	0
adaptive	pointer	1e+04	0.4743	[0.467, 0.482]	0.00988	5	0
adaptive	seq2seq	0	0.4363	[0.427, 0.444]	0.0112	5	0
adaptive	seq2seq	0.01	0.4345	[0.425, 0.443]	0.0118	5	0
adaptive	seq2seq	0.1	0.4336	[0.425, 0.442]	0.0111	5	0
adaptive	seq2seq	1	0.4344	[0.425, 0.443]	0.0107	5	0
adaptive	seq2seq	10	0.4338	[0.425, 0.441]	0.0105	5	0
adaptive	seq2seq	100	0.4361	[0.427, 0.445]	0.0118	5	0
adaptive	seq2seq	1000	0.4361	[0.427, 0.446]	0.0123	5	0
adaptive	seq2seq	1e+04	0.4296	[0.426, 0.434]	0.00566	5	0
tropical	pointer	0	0.494	[0.474, 0.514]	0.0262	5	0
tropical	pointer	0.01	0.4929	[0.474, 0.514]	0.0258	5	0
tropical	pointer	0.1	0.4896	[0.471, 0.508]	0.0248	5	0
tropical	pointer	1	0.495	[0.475, 0.517]	0.03	5	0
tropical	pointer	10	0.4852	[0.47, 0.503]	0.0214	5	0
tropical	pointer	100	0.4932	[0.474, 0.515]	0.0265	5	0
tropical	pointer	1000	0.4875	[0.472, 0.509]	0.0253	5	0
tropical	pointer	1e+04	0.4948	[0.479, 0.511]	0.0212	5	0
tropical	seq2seq	0	0.4586	[0.437, 0.486]	0.0317	5	0
tropical	seq2seq	0.01	0.475	[0.457, 0.495]	0.0265	5	0
tropical	seq2seq	0.1	0.4781	[0.461, 0.5]	0.025	5	0
tropical	seq2seq	1	0.4649	[0.444, 0.489]	0.0287	5	0
tropical	seq2seq	10	0.4672	[0.448, 0.491]	0.028	5	0
tropical	seq2seq	100	0.459	[0.436, 0.485]	0.0313	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1000	0.4439	[0.412, 0.484]	0.0475	5	0
tropical	seq2seq	1e+04	0.4248	[0.405, 0.446]	0.0251	5	0
vanilla	pointer	0	0.4739	[0.468, 0.479]	0.00705	5	0
vanilla	pointer	0.01	0.4718	[0.466, 0.478]	0.00761	5	0
vanilla	pointer	0.1	0.4728	[0.468, 0.478]	0.00653	5	0
vanilla	pointer	1	0.474	[0.47, 0.478]	0.00479	5	0
vanilla	pointer	10	0.4731	[0.468, 0.478]	0.00606	5	0
vanilla	pointer	100	0.4732	[0.469, 0.477]	0.00525	5	0
vanilla	pointer	1000	0.474	[0.469, 0.479]	0.00619	5	0
vanilla	pointer	1e+04	0.4748	[0.472, 0.477]	0.00368	5	0
vanilla	seq2seq	0	0.4568	[0.446, 0.468]	0.0137	5	0
vanilla	seq2seq	0.01	0.4567	[0.451, 0.463]	0.00763	5	0
vanilla	seq2seq	0.1	0.4544	[0.45, 0.459]	0.00544	5	0
vanilla	seq2seq	1	0.4565	[0.453, 0.462]	0.00596	5	0
vanilla	seq2seq	10	0.4601	[0.455, 0.466]	0.0075	5	0
vanilla	seq2seq	100	0.462	[0.455, 0.469]	0.00874	5	0
vanilla	seq2seq	1000	0.4576	[0.446, 0.467]	0.0139	5	0
vanilla	seq2seq	1e+04	0.4269	[0.419, 0.434]	0.00985	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.001966	[-0.0147, 0.0186]	0.9997
adaptive	pointer	0.1	7.002×10^{-4}	[-0.0159, 0.0173]	1.0000
adaptive	pointer	1	0.00205	[-0.0146, 0.0187]	0.9996
adaptive	pointer	10	0.001234	[-0.0154, 0.0179]	1.0000
adaptive	pointer	100	0.001936	[-0.0147, 0.0186]	0.9997
adaptive	pointer	1000	0.00251	[-0.0141, 0.0191]	0.9985
adaptive	pointer	1e+04	0.001455	[-0.0152, 0.0181]	1.0000
adaptive	seq2seq	0.01	-0.001755	[-0.0206, 0.0171]	0.9999
adaptive	seq2seq	0.1	-0.002651	[-0.0215, 0.0162]	0.9990
adaptive	seq2seq	1	-0.001922	[-0.0208, 0.017]	0.9999
adaptive	seq2seq	10	-0.00247	[-0.0213, 0.0164]	0.9994
adaptive	seq2seq	100	-1.977×10^{-4}	[-0.0191, 0.0187]	1.0000
adaptive	seq2seq	1000	-2.062×10^{-4}	[-0.0191, 0.0187]	1.0000
adaptive	seq2seq	1e+04	-0.006635	[-0.0255, 0.0122]	0.8702
tropical	pointer	0.01	-0.001179	[-0.0454, 0.043]	1.0000
tropical	pointer	0.1	-0.004494	[-0.0487, 0.0397]	0.9999
tropical	pointer	1	9.210×10^{-4}	[-0.0433, 0.0451]	1.0000
tropical	pointer	10	-0.008895	[-0.0531, 0.0353]	0.9914
tropical	pointer	100	-8.480×10^{-4}	[-0.045, 0.0433]	1.0000
tropical	pointer	1000	-0.006517	[-0.0507, 0.0377]	0.9987
tropical	pointer	1e+04	7.178×10^{-4}	[-0.0435, 0.0449]	1.0000
tropical	seq2seq	0.01	0.01644	[-0.0381, 0.071]	0.9321
tropical	seq2seq	0.1	0.0195	[-0.0351, 0.0741]	0.8617
tropical	seq2seq	1	0.006372	[-0.0482, 0.0609]	0.9997
tropical	seq2seq	10	0.008602	[-0.046, 0.0632]	0.9980
tropical	seq2seq	100	4.860×10^{-4}	[-0.0541, 0.0551]	1.0000
tropical	seq2seq	1000	-0.01468	[-0.0693, 0.0399]	0.9600
tropical	seq2seq	1e+04	-0.03377	[-0.0883, 0.0208]	0.3845
vanilla	pointer	0.01	-0.002072	[-0.0126, 0.00843]	0.9924
vanilla	pointer	0.1	-0.001092	[-0.0116, 0.00941]	0.9999
vanilla	pointer	1	1.152×10^{-4}	[-0.0104, 0.0106]	1.0000

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
vanilla	pointer	10	-7.854×10^{-4}	[-0.0113, 0.00972]	1.0000	
vanilla	pointer	100	-6.320×10^{-4}	[-0.0111, 0.00987]	1.0000	
vanilla	pointer	1000	1.260×10^{-4}	[-0.0104, 0.0106]	1.0000	
vanilla	pointer	1e+04	9.510×10^{-4}	[-0.00955, 0.0115]	0.9999	
vanilla	seq2seq	0.01	-6.422×10^{-5}	[-0.0168, 0.0167]	1.0000	
vanilla	seq2seq	0.1	-0.002382	[-0.0191, 0.0144]	0.9989	
vanilla	seq2seq	1	-2.292×10^{-4}	[-0.017, 0.0165]	1.0000	
vanilla	seq2seq	10	0.00336	[-0.0134, 0.0201]	0.9916	
vanilla	seq2seq	100	0.005209	[-0.0115, 0.0219]	0.9215	
vanilla	seq2seq	1000	7.980×10^{-4}	[-0.0159, 0.0175]	1.0000	
vanilla	seq2seq	1e+04	-0.02983	[-0.0466, -0.0131]	0.0002	*

Table 3: `ood_len_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	7.052×10^{-5}	$[-1.45 \times 10^{-4}, 2.69 \times 10^{-4}]$	0.9227	0.0514	0.7031
adaptive	seq2seq	-2.867×10^{-4}	[-0.00158, 9.51×10^{-4}]	0.7392	-0.0362	0.8090
tropical	pointer	-4.458×10^{-6}	[-0.00113, 0.00123]	0.9982	0.0133	0.9312
tropical	seq2seq	-0.008031	[-0.0106, -0.00582]	0.0029	-0.406	0.0020
vanilla	pointer	3.842×10^{-4}	$[-5.00 \times 10^{-5}, 8.47 \times 10^{-4}]$	0.4071	0.0781	0.5735
vanilla	seq2seq	-0.002768	[-0.00451, -0.00103]	0.0152	-0.101	0.4489

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.02019	[-0.0132, 0.0536]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.02108	[-0.0152, 0.0574]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.01679	[-0.0173, 0.0508]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.021	[-0.0183, 0.0603]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.01208	[-0.016, 0.0402]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.01998	[-0.0146, 0.0545]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.01355	[-0.0237, 0.0508]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.01996	[-0.0094, 0.0493]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.001784	[-0.0369, 0.0405]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.01829	[-0.0162, 0.0527]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.02366	[-0.0108, 0.0581]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.008385	[-0.0332, 0.0499]	1.0000

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.007026	[-0.0328, 0.0469]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.002939	[-0.0383, 0.0324]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.0137	[-0.0678, 0.0404]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.002155	[-0.0247, 0.0204]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.001006	[-0.0199, 0.0178]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.003031	[-0.014, 0.02]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	7.865×10^{-4}	[-0.0166, 0.0182]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	9.283×10^{-4}	[-0.0138, 0.0157]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.001014	[-0.0157, 0.0178]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.001562	[-0.0138, 0.0169]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.001378	[-0.0154, 0.0181]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-5.025×10^{-4}	[-0.0163, 0.0152]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.0205	[-0.0481, 0.00709]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.02219	[-0.0421, -0.00232]	0.5068	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.02077	[-0.0396, -0.00194]	0.5068	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.02219	[-0.0387, -0.00569]	0.3031	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.02633	[-0.0442, -0.00845]	0.2397	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.0259	[-0.0503, -0.00147]	0.5071	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.0215	[-0.0474, 0.0044]	0.9075	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.002698	[-0.0145, 0.0198]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.03657	[0.0116, 0.0615]	0.1984	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.04029	[0.0201, 0.0605]	0.0987	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.03992	[0.0198, 0.06]	0.0987	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.04054	[0.0216, 0.0595]	0.0806	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.04028	[0.0189, 0.0616]	0.1076	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.03871	[0.0162, 0.0612]	0.1417	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.03929	[0.0151, 0.0634]	0.1603	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.04466	[0.0293, 0.0601]	0.0297	*

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.03549	[0.00817, 0.0628]	0.2716	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.01787	[-0.0233, 0.0591]	0.7449	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.0115	[-0.0281, 0.0511]	0.7449	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.03004	[-0.0179, 0.078]	0.7449	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.01799	[-0.00956, 0.0455]	0.7449	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.03415	$[-4.51 \times 10^{-4}, 0.0688]$	0.4670	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.04365	[-0.017, 0.104]	0.7449	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.06998	[0.0426, 0.0973]	0.0457	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.01708	[-0.00452, 0.0387]	0.7449	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.01507	$[7.39 \times 10^{-4}, 0.0294]$	0.4657	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.01837	[0.00631, 0.0304]	0.1872	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.01742	[0.00995, 0.0249]	0.0616	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.01293	$[7.25 \times 10^{-4}, 0.0251]$	0.4657	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.01124	[-0.00545, 0.0279]	0.7449	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.01641	[-0.00532, 0.0381]	0.7449	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.04786	[0.0375, 0.0582]	0.0051	*

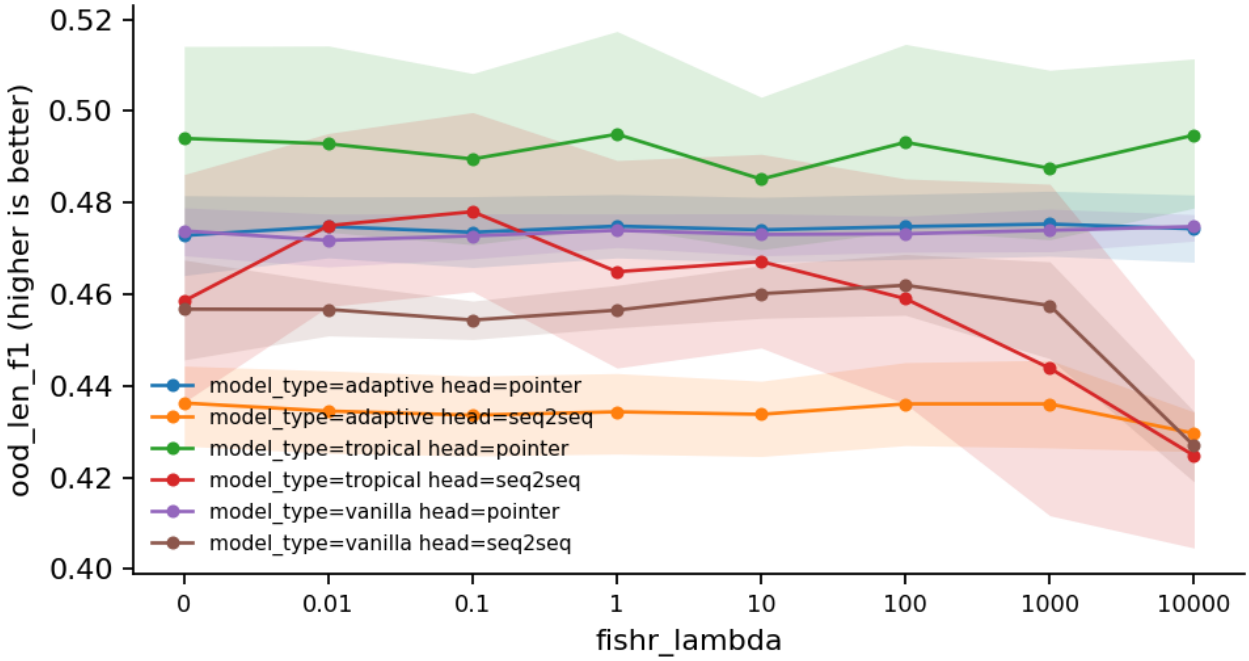


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

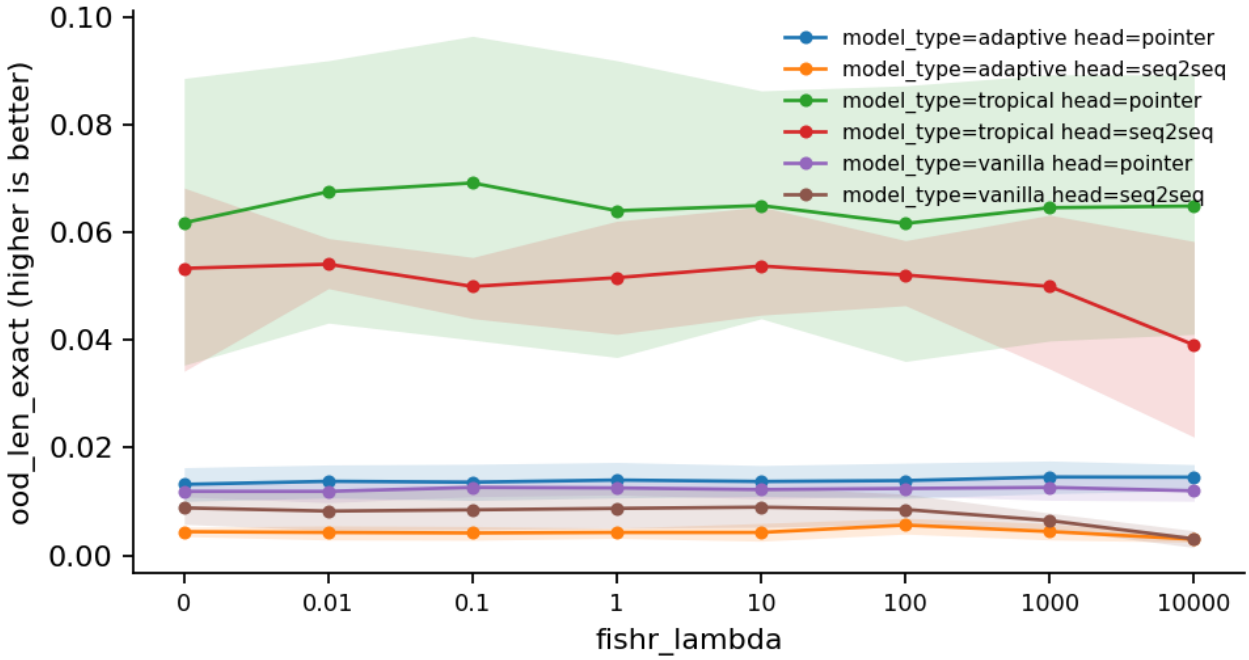


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.0132	[0.0101, 0.0163]	0.00408	5	0
adaptive	pointer	0.01	0.01377	[0.0107, 0.0168]	0.00385	5	0
adaptive	pointer	0.1	0.0136	[0.0102, 0.017]	0.00424	5	0
adaptive	pointer	1	0.014	[0.0107, 0.0173]	0.00425	5	0
adaptive	pointer	10	0.01373	[0.011, 0.0167]	0.00378	5	0
adaptive	pointer	100	0.0139	[0.0106, 0.0172]	0.00411	5	0
adaptive	pointer	1000	0.01457	[0.0115, 0.0176]	0.00403	5	0
adaptive	pointer	1e+04	0.01453	[0.0121, 0.0169]	0.0032	5	0
adaptive	seq2seq	0	0.0044	[0.00353, 0.00527]	0.00113	5	0
adaptive	seq2seq	0.01	0.004267	[0.0029, 0.0055]	0.00173	5	0
adaptive	seq2seq	0.1	0.004167	[0.00283, 0.0054]	0.00163	5	0
adaptive	seq2seq	1	0.004267	[0.0032, 0.0051]	0.00118	5	0
adaptive	seq2seq	10	0.004267	[0.00267, 0.00597]	0.00208	5	0
adaptive	seq2seq	100	0.005633	[0.004, 0.00693]	0.00182	5	0
adaptive	seq2seq	1000	0.004433	[0.0029, 0.00593]	0.00191	5	0
adaptive	seq2seq	1e+04	0.003	[0.00247, 0.00363]	7.26×10^{-4}	5	0
tropical	pointer	0	0.0618	[0.0354, 0.0886]	0.0352	5	0
tropical	pointer	0.01	0.06757	[0.0432, 0.0919]	0.0319	5	0
tropical	pointer	0.1	0.0692	[0.04, 0.0965]	0.037	5	0
tropical	pointer	1	0.06403	[0.0368, 0.0919]	0.035	5	0
tropical	pointer	10	0.065	[0.044, 0.0863]	0.0275	5	0
tropical	pointer	100	0.06167	[0.0361, 0.0873]	0.033	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	0.0646	[0.0399, 0.0893]	0.0321	5	0
tropical	pointer	1e+04	0.0649	[0.0411, 0.0893]	0.0312	5	0
tropical	seq2seq	0	0.05333	[0.0342, 0.0683]	0.0217	5	0
tropical	seq2seq	0.01	0.0541	[0.0496, 0.0589]	0.00602	5	0
tropical	seq2seq	0.1	0.04997	[0.044, 0.0554]	0.00747	5	0
tropical	seq2seq	1	0.0516	[0.0411, 0.0621]	0.014	5	0
tropical	seq2seq	10	0.05377	[0.0447, 0.0648]	0.0128	5	0
tropical	seq2seq	100	0.0521	[0.0464, 0.0585]	0.00804	5	0
tropical	seq2seq	1000	0.04997	[0.0347, 0.0632]	0.0173	5	0
tropical	seq2seq	1e+04	0.0391	[0.022, 0.0583]	0.0226	5	0
vanilla	pointer	0	0.0119	[0.0109, 0.0131]	0.00132	5	0
vanilla	pointer	0.01	0.0119	[0.00983, 0.0139]	0.00263	5	0
vanilla	pointer	0.1	0.01263	[0.0109, 0.0142]	0.00215	5	0
vanilla	pointer	1	0.01253	[0.0113, 0.0138]	0.00163	5	0
vanilla	pointer	10	0.01223	[0.0105, 0.0137]	0.00207	5	0
vanilla	pointer	100	0.01243	[0.0111, 0.0136]	0.00171	5	0
vanilla	pointer	1000	0.01263	[0.0103, 0.015]	0.00311	5	0
vanilla	pointer	1e+04	0.012	[0.0101, 0.0142]	0.00268	5	0
vanilla	seq2seq	0	0.008833	[0.0058, 0.0121]	0.00407	5	0
vanilla	seq2seq	0.01	0.008233	[0.0047, 0.0118]	0.00472	5	0
vanilla	seq2seq	0.1	0.008467	[0.0048, 0.0128]	0.00512	5	0
vanilla	seq2seq	1	0.008733	[0.00507, 0.0124]	0.00483	5	0
vanilla	seq2seq	10	0.008967	[0.00523, 0.0127]	0.00479	5	0
vanilla	seq2seq	100	0.008533	[0.0058, 0.0114]	0.00369	5	0
vanilla	seq2seq	1000	0.006467	[0.00493, 0.00787]	0.00186	5	0
vanilla	seq2seq	1e+04	0.003067	[0.0015, 0.0046]	0.00192	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.001167	$[-5.00 \times 10^{-4}, 0.00123]$	1.0000
adaptive	pointer	0.1	1.667×10^{-4}	$[-7.00 \times 10^{-4}, 0.0015]$	1.0000
adaptive	pointer	1	5.000×10^{-4}	$[3.00 \times 10^{-4}, 0.0014]$	0.4375
adaptive	pointer	10	8.333×10^{-4}	$[-4.67 \times 10^{-4}, 0.00137]$	0.9375
adaptive	pointer	100	8.333×10^{-4}	$[3.33 \times 10^{-5}, 0.00134]$	0.7500
adaptive	pointer	1000	0.001833	$[4.67 \times 10^{-4}, 0.00207]$	0.7500
adaptive	pointer	1e+04	0.001167	$[6.67 \times 10^{-5}, 0.00263]$	0.7500
adaptive	seq2seq	0.01	-5.000×10^{-4}	$[-6.33 \times 10^{-4}, 5.33 \times 10^{-4}]$	1.0000
adaptive	seq2seq	0.1	-1.667×10^{-4}	$[-6.33 \times 10^{-4}, 2.00 \times 10^{-4}]$	1.0000
adaptive	seq2seq	1	-1.667×10^{-4}	$[-2.33 \times 10^{-4}, -3.33 \times 10^{-5}]$	1.0000
adaptive	seq2seq	10	-6.667×10^{-4}	$[-8.00 \times 10^{-4}, 9.67 \times 10^{-4}]$	1.0000
adaptive	seq2seq	100	0.001333	$[4.67 \times 10^{-4}, 0.002]$	0.8750
adaptive	seq2seq	1000	0	$[-6.67 \times 10^{-4}, 9.00 \times 10^{-4}]$	1.0000
adaptive	seq2seq	1e+04	-0.001333	$[-0.00233, -5.00 \times 10^{-4}]$	0.8750
tropical	pointer	0.01	0.006167	[-0.00117, 0.0133]	1.0000
tropical	pointer	0.1	0.01133	[-0.00307, 0.0179]	1.0000
tropical	pointer	1	-0.002833	[-0.00963, 0.0179]	1.0000
tropical	pointer	10	0.0025	[-0.00707, 0.0148]	1.0000
tropical	pointer	100	-1.667×10^{-4}	[-0.00644, 0.00753]	1.0000
tropical	pointer	1000	0.006833	[-0.0068, 0.0127]	1.0000
tropical	pointer	1e+04	0.004667	[-0.0095, 0.014]	1.0000
tropical	seq2seq	0.01	-0.006667	[-0.0127, 0.0166]	1.0000

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
tropical	seq2seq	0.1	-0.01333	[-0.0174, 0.0139]	1.0000
tropical	seq2seq	1	-0.0065	[-0.0161, 0.0145]	1.0000
tropical	seq2seq	10	-0.004167	[-0.0131, 0.0176]	1.0000
tropical	seq2seq	100	-0.009	[-0.0117, 0.0122]	1.0000
tropical	seq2seq	1000	-0.005333	[-0.00673, 8.67×10^{-4}]	1.0000
tropical	seq2seq	1e+04	-0.01617	[-0.021, -0.00743]	0.4375
vanilla	pointer	0.01	1.667×10^{-4}	[-0.0016, 0.00153]	1.0000
vanilla	pointer	0.1	1.000×10^{-3}	[-2.67×10^{-4}, 0.0018]	1.0000
vanilla	pointer	1	3.333×10^{-4}	[-2.67×10^{-4}, 0.00177]	1.0000
vanilla	pointer	10	8.333×10^{-4}	[-0.00103, 0.00127]	1.0000
vanilla	pointer	100	0.001333	[-8.37×10^{-4}, 0.00187]	1.0000
vanilla	pointer	1000	0.001833	[-9.00×10^{-4}, 0.00233]	1.0000
vanilla	pointer	1e+04	-1.667×10^{-4}	[-0.00117, 0.00143]	1.0000
vanilla	seq2seq	0.01	-0.001333	[-0.00377, 0.00337]	1.0000
vanilla	seq2seq	0.1	-0.0015	[-0.00357, 0.00393]	1.0000
vanilla	seq2seq	1	-8.333×10^{-4}	[-0.00323, 0.00387]	1.0000
vanilla	seq2seq	10	-0.001	[-0.00187, 0.0032]	1.0000
vanilla	seq2seq	100	-3.333×10^{-4}	[-0.00113, 5.33×10^{-4}]	1.0000
vanilla	seq2seq	1000	-0.001667	[-0.00553, 9.33×10^{-4}]	1.0000
vanilla	seq2seq	1e+04	-0.003	[-0.00947, -0.00247]	0.4375

Table 7: `ood_len_exact` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.05381	[-0.258, 0.32]	0.7588	0.0438	0.7375
adaptive	seq2seq	-0.131	[-0.331, 0.0872]	0.4530	-0.0971	0.4781
tropical	pointer	-0.058	[-0.244, 0.0815]	0.7407	-0.0362	0.7862
tropical	seq2seq	-0.198	[-0.557, 0.186]	0.2541	-0.166	0.2282
vanilla	pointer	-0.02336	[-0.151, 0.156]	0.8940	-0.00762	0.9628
vanilla	seq2seq	-0.3411	[-0.688, 0.0277]	0.0449	-0.284	0.0312

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0499	[0.00562, 0.0942]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.05567	[0.0163, 0.0951]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.05657	[0.0111, 0.102]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.0515	[0.0084, 0.0946]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.05277	[0.0195, 0.086]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.04923	[0.00859, 0.0899]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.05197	[0.0119, 0.092]	1.0000

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.0529	[0.0141, 0.0917]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.0445	[0.02, 0.069]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.04587	[0.0375, 0.0542]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.0415	[0.0317, 0.0513]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.04287	[0.0267, 0.059]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.0448	[0.0298, 0.0598]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.04357	[0.0342, 0.0529]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.0435	[0.0212, 0.0658]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.03603	[0.00969, 0.0624]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.0013	[-0.00405, 0.00665]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.001867	[-0.00345, 0.00719]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	9.667×10^{-4}	[-0.00461, 0.00655]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.001467	[-0.00388, 0.00682]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.0015	[-0.00316, 0.00616]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.001467	[-0.00332, 0.00625]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.001933	[-0.00419, 0.00806]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.002533	[-0.00202, 0.00708]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.004433	[-0.0104, 0.00155]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.003967	[-0.00967, 0.00173]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.0043	[-0.0106, 0.00203]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.004467	[-0.0107, 0.00178]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.0047	[-0.0104, 9.96×10^{-4}]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.0029	[-0.00779, 0.00199]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.002033	[-0.00413, 6.39×10^{-5}]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-6.667×10^{-5}	[-0.00189, 0.00175]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.0088	[0.00268, 0.0149]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.0095	[0.00344, 0.0156]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.009433	[0.00288, 0.016]	1.0000

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.009733	[0.00344, 0.016]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.009467	[0.00287, 0.0161]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.008267	[0.00202, 0.0145]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.01013	[0.00322, 0.017]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.01153	[0.00746, 0.0156]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.008467	[-0.0399, 0.0568]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.01347	[-0.027, 0.0539]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.01923	[-0.0317, 0.0702]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.01243	[-0.038, 0.0629]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.01123	[-0.0321, 0.0545]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.009567	[-0.0342, 0.0533]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.01463	[-0.0246, 0.0539]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.0258	[-0.0261, 0.0777]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.003067	[-0.00273, 0.00886]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.003667	[-0.002, 0.00933]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.004167	[-0.00249, 0.0108]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.0038	[-0.00234, 0.00994]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.003267	[-0.0026, 0.00913]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.0039	$[-7.01 \times 10^{-4}, 0.0085]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.006167	$[1.10 \times 10^{-4}, 0.0122]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.008933	[0.00425, 0.0136]	1.0000

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.7275	[0.709, 0.746]	0.0245	5	0
adaptive	pointer	0.01	0.7298	[0.712, 0.748]	0.0224	5	0
adaptive	pointer	0.1	0.7225	[0.701, 0.742]	0.0281	5	0
adaptive	pointer	1	0.7297	[0.719, 0.74]	0.0143	5	0
adaptive	pointer	10	0.7279	[0.717, 0.739]	0.0148	5	0
adaptive	pointer	100	0.7241	[0.707, 0.741]	0.0217	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	1000	0.7245	[0.712, 0.739]	0.0168	5	0
adaptive	pointer	1e+04	0.7239	[0.707, 0.737]	0.0192	5	0
adaptive	seq2seq	0	0.6824	[0.66, 0.703]	0.0292	5	0
adaptive	seq2seq	0.01	0.6723	[0.641, 0.695]	0.0366	5	0
adaptive	seq2seq	0.1	0.6703	[0.646, 0.69]	0.0286	5	0
adaptive	seq2seq	1	0.6701	[0.649, 0.688]	0.0243	5	0
adaptive	seq2seq	10	0.6885	[0.667, 0.705]	0.0258	5	0
adaptive	seq2seq	100	0.6804	[0.657, 0.706]	0.0322	5	0
adaptive	seq2seq	1000	0.6729	[0.632, 0.713]	0.052	5	0
adaptive	seq2seq	1e+04	0.6499	[0.612, 0.684]	0.0473	5	0
tropical	pointer	0	0.6735	[0.629, 0.716]	0.058	5	0
tropical	pointer	0.01	0.6096	[0.574, 0.645]	0.0476	5	0
tropical	pointer	0.1	0.6188	[0.569, 0.668]	0.0623	5	0
tropical	pointer	1	0.6575	[0.63, 0.687]	0.0384	5	0
tropical	pointer	10	0.6412	[0.608, 0.672]	0.0419	5	0
tropical	pointer	100	0.6219	[0.543, 0.692]	0.0974	5	0
tropical	pointer	1000	0.6436	[0.584, 0.695]	0.0737	5	0
tropical	pointer	1e+04	0.6425	[0.581, 0.691]	0.0649	5	0
tropical	seq2seq	0	0.6178	[0.469, 0.706]	0.168	5	0
tropical	seq2seq	0.01	0.6529	[0.604, 0.698]	0.0593	5	0
tropical	seq2seq	0.1	0.6535	[0.618, 0.693]	0.0488	5	0
tropical	seq2seq	1	0.5923	[0.532, 0.65]	0.079	5	0
tropical	seq2seq	10	0.6031	[0.56, 0.633]	0.0471	5	0
tropical	seq2seq	100	0.6231	[0.506, 0.693]	0.13	5	0
tropical	seq2seq	1000	0.5732	[0.445, 0.691]	0.158	5	0
tropical	seq2seq	1e+04	0.5283	[0.447, 0.617]	0.108	5	0
vanilla	pointer	0	0.6541	[0.594, 0.705]	0.0691	5	0
vanilla	pointer	0.01	0.6321	[0.55, 0.714]	0.105	5	0
vanilla	pointer	0.1	0.6298	[0.536, 0.719]	0.116	5	0
vanilla	pointer	1	0.6235	[0.532, 0.715]	0.115	5	0
vanilla	pointer	10	0.634	[0.542, 0.711]	0.105	5	0
vanilla	pointer	100	0.6268	[0.547, 0.7]	0.0985	5	0
vanilla	pointer	1000	0.6147	[0.523, 0.706]	0.116	5	0
vanilla	pointer	1e+04	0.6304	[0.549, 0.712]	0.104	5	0
vanilla	seq2seq	0	0.6565	[0.622, 0.701]	0.0512	5	0
vanilla	seq2seq	0.01	0.6764	[0.635, 0.716]	0.0535	5	0
vanilla	seq2seq	0.1	0.6816	[0.643, 0.72]	0.0495	5	0
vanilla	seq2seq	1	0.6779	[0.639, 0.715]	0.0493	5	0
vanilla	seq2seq	10	0.6752	[0.641, 0.706]	0.0418	5	0
vanilla	seq2seq	100	0.6621	[0.633, 0.697]	0.0415	5	0
vanilla	seq2seq	1000	0.6303	[0.574, 0.674]	0.0676	5	0
vanilla	seq2seq	1e+04	0.6519	[0.609, 0.694]	0.0551	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.002339	[-0.0339, 0.0385]	1.0000
adaptive	pointer	0.1	-0.005037	[-0.0412, 0.0312]	0.9991
adaptive	pointer	1	0.002188	[-0.034, 0.0384]	1.0000
adaptive	pointer	10	4.367×10^{-4}	[-0.0358, 0.0366]	1.0000
adaptive	pointer	100	-0.003417	[-0.0396, 0.0328]	0.9999
adaptive	pointer	1000	-0.003015	[-0.0392, 0.0332]	1.0000
adaptive	pointer	1e+04	-0.003587	[-0.0398, 0.0326]	0.9999

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
adaptive	seq2seq	0.01	-0.01006	[-0.0726, 0.0524]	0.9977
adaptive	seq2seq	0.1	-0.01203	[-0.0745, 0.0505]	0.9934
adaptive	seq2seq	1	-0.01224	[-0.0747, 0.0503]	0.9927
adaptive	seq2seq	10	0.006134	[-0.0564, 0.0686]	0.9999
adaptive	seq2seq	100	-0.001954	[-0.0645, 0.0605]	1.0000
adaptive	seq2seq	1000	-0.009445	[-0.0719, 0.0531]	0.9984
adaptive	seq2seq	1e+04	-0.03243	[-0.0949, 0.0301]	0.5644
tropical	pointer	0.01	-0.06392	[-0.174, 0.0463]	0.4509
tropical	pointer	0.1	-0.05472	[-0.165, 0.0555]	0.6086
tropical	pointer	1	-0.01603	[-0.126, 0.0942]	0.9988
tropical	pointer	10	-0.03235	[-0.143, 0.0779]	0.9398
tropical	pointer	100	-0.05167	[-0.162, 0.0586]	0.6625
tropical	pointer	1000	-0.02989	[-0.14, 0.0804]	0.9585
tropical	pointer	1e+04	-0.03104	[-0.141, 0.0792]	0.9503
tropical	seq2seq	0.01	0.03507	[-0.156, 0.227]	0.9950
tropical	seq2seq	0.1	0.0357	[-0.156, 0.227]	0.9945
tropical	seq2seq	1	-0.0255	[-0.217, 0.166]	0.9993
tropical	seq2seq	10	-0.01473	[-0.206, 0.177]	1.0000
tropical	seq2seq	100	0.005224	[-0.186, 0.197]	1.0000
tropical	seq2seq	1000	-0.0446	[-0.236, 0.147]	0.9808
tropical	seq2seq	1e+04	-0.08949	[-0.281, 0.102]	0.6652
vanilla	pointer	0.01	-0.02196	[-0.205, 0.161]	0.9996
vanilla	pointer	0.1	-0.0243	[-0.207, 0.158]	0.9993
vanilla	pointer	1	-0.03056	[-0.213, 0.152]	0.9971
vanilla	pointer	10	-0.0201	[-0.203, 0.163]	0.9998
vanilla	pointer	100	-0.02729	[-0.21, 0.155]	0.9985
vanilla	pointer	1000	-0.03935	[-0.222, 0.143]	0.9874
vanilla	pointer	1e+04	-0.02368	[-0.206, 0.159]	0.9994
vanilla	seq2seq	0.01	0.0199	[-0.0706, 0.11]	0.9859
vanilla	seq2seq	0.1	0.02511	[-0.0653, 0.116]	0.9536
vanilla	seq2seq	1	0.02138	[-0.0691, 0.112]	0.9793
vanilla	seq2seq	10	0.01866	[-0.0718, 0.109]	0.9901
vanilla	seq2seq	100	0.00555	[-0.0849, 0.096]	1.0000
vanilla	seq2seq	1000	-0.02624	[-0.117, 0.0642]	0.9428
vanilla	seq2seq	1e+04	-0.004593	[-0.095, 0.0859]	1.0000

Table 11: `ood_val_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-6.907×10^{-4}	[-0.00175, 6.53×10^{-4}]	0.6650	-0.0629	0.6371
adaptive	seq2seq	-0.001845	[-0.0105, 0.00867]	0.5412	-0.0438	0.7632
tropical	pointer	0.004023	[-0.00338, 0.0106]	0.4363	0.15	0.2649
tropical	seq2seq	-0.01798	[-0.0343, -0.00171]	0.0289	-0.189	0.1554
vanilla	pointer	-0.001142	[-0.00352, 0.00166]	0.8936	-0.0362	0.7930
vanilla	seq2seq	-0.006857	[-0.0123, -0.00144]	0.1069	-0.177	0.1850

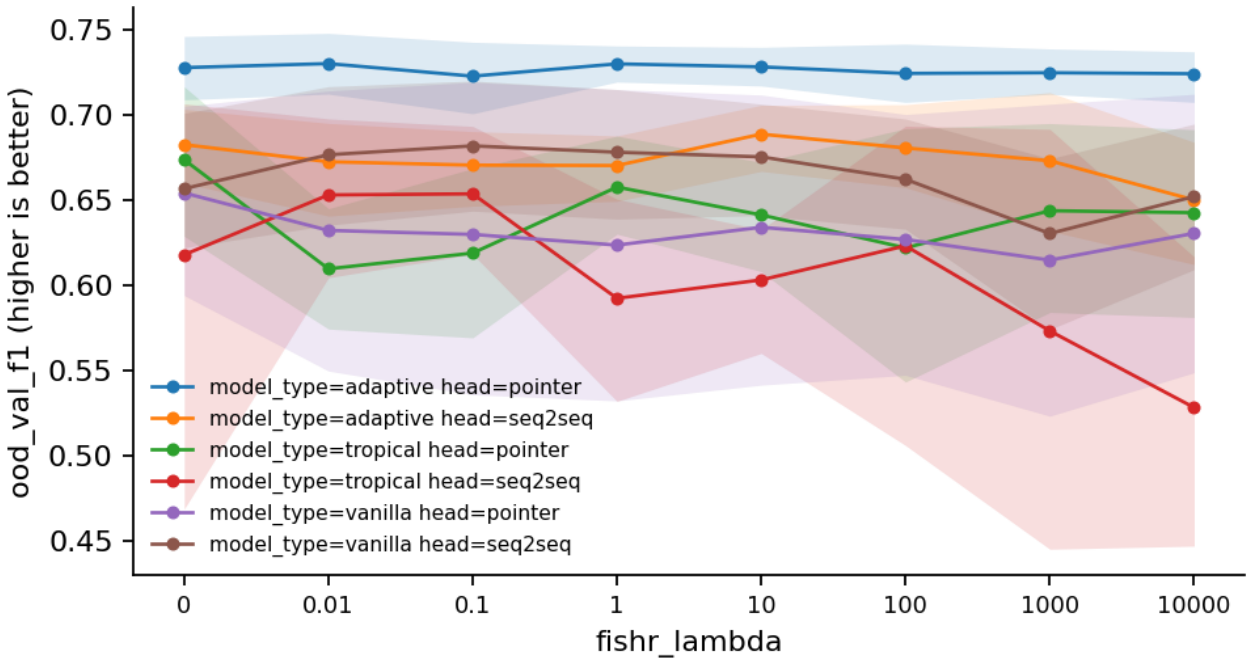


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.01946	[-0.0726, 0.112]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.0225	[-0.189, 0.144]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.01095	[-0.171, 0.149]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.03399	[-0.112, 0.18]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.00722	[-0.102, 0.116]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.004914	[-0.16, 0.15]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.02893	[-0.124, 0.181]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.0121	[-0.133, 0.157]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.0387	[-0.212, 0.135]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.02353	[-0.123, 0.0763]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.02811	[-0.124, 0.0679]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.08558	[-0.195, 0.0234]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.07209	[-0.149, 0.00445]	0.9251

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.03902	[-0.189, 0.111]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.05706	[-0.222, 0.108]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.1236	[-0.254, 0.00658]	0.9251
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.07344	[-0.0176, 0.164]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.09774	[-0.0312, 0.227]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.0927	[-0.053, 0.238]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.1062	[-0.0383, 0.251]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.09398	[-0.0324, 0.22]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.09732	[-0.0212, 0.216]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.1098	[-0.0311, 0.251]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.09353	[-0.0397, 0.227]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.02584	[-0.0448, 0.0965]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.004114	[-0.101, 0.0927]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01129	[-0.0887, 0.0661]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.007781	[-0.088, 0.0725]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01332	[-0.0226, 0.0492]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.01834	[-0.0177, 0.0543]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.04264	[-0.0353, 0.121]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.001997	[-0.0517, 0.0477]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.04513	[0.00707, 0.0832]	0.6934
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.05753	[0.00845, 0.107]	0.6934
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.05212	[-0.00209, 0.106]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.05956	[0.0239, 0.0953]	0.2352
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.03943	[-0.00848, 0.0873]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.04367	[-0.005, 0.0923]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.05156	[-0.0254, 0.129]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.07397	[0.0061, 0.142]	0.8177
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.05569	[-0.15, 0.261]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.0433	[-0.162, 0.0755]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.03472	[-0.156, 0.0866]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.06517	[-0.0483, 0.179]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.03808	[-0.0342, 0.11]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.001205	[-0.252, 0.25]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.07041	[-0.149, 0.29]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.1141	[-0.0805, 0.309]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.002468	[-0.0413, 0.0364]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.04433	[-0.18, 0.0912]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.05188	[-0.204, 0.1]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.0544	[-0.202, 0.0928]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.04123	[-0.184, 0.102]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.03531	[-0.146, 0.0755]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.01558	[-0.155, 0.124]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.02155	[-0.176, 0.132]	1.0000

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.2621	[0.233, 0.287]	0.035	5	0
adaptive	pointer	0.01	0.2542	[0.231, 0.278]	0.032	5	0
adaptive	pointer	0.1	0.2457	[0.217, 0.274]	0.0374	5	0
adaptive	pointer	1	0.2566	[0.246, 0.268]	0.0144	5	0
adaptive	pointer	10	0.251	[0.238, 0.265]	0.0178	5	0
adaptive	pointer	100	0.2474	[0.222, 0.271]	0.0316	5	0
adaptive	pointer	1000	0.2452	[0.227, 0.265]	0.0238	5	0
adaptive	pointer	1e+04	0.2411	[0.213, 0.267]	0.033	5	0
adaptive	seq2seq	0	0.1268	[0.0989, 0.149]	0.0326	5	0
adaptive	seq2seq	0.01	0.1226	[0.102, 0.142]	0.0258	5	0
adaptive	seq2seq	0.1	0.1173	[0.1, 0.132]	0.0204	5	0
adaptive	seq2seq	1	0.1106	[0.0898, 0.13]	0.0237	5	0
adaptive	seq2seq	10	0.1337	[0.106, 0.158]	0.0319	5	0
adaptive	seq2seq	100	0.1424	[0.112, 0.172]	0.0398	5	0
adaptive	seq2seq	1000	0.1394	[0.115, 0.164]	0.0325	5	0
adaptive	seq2seq	1e+04	0.1137	[0.0934, 0.13]	0.0234	5	0
tropical	pointer	0	0.1947	[0.138, 0.25]	0.0709	5	0
tropical	pointer	0.01	0.1582	[0.12, 0.202]	0.0524	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0.1	0.1654	[0.104, 0.227]	0.0787	5	0
tropical	pointer	1	0.1851	[0.152, 0.218]	0.045	5	0
tropical	pointer	10	0.1924	[0.144, 0.232]	0.058	5	0
tropical	pointer	100	0.1714	[0.112, 0.231]	0.0758	5	0
tropical	pointer	1000	0.1791	[0.118, 0.244]	0.0784	5	0
tropical	pointer	1e+04	0.1792	[0.133, 0.23]	0.0637	5	0
tropical	seq2seq	0	0.09906	[0.0653, 0.136]	0.0461	5	0
tropical	seq2seq	0.01	0.09706	[0.055, 0.139]	0.0538	5	0
tropical	seq2seq	0.1	0.09208	[0.0569, 0.127]	0.0453	5	0
tropical	seq2seq	1	0.09226	[0.0571, 0.126]	0.0452	5	0
tropical	seq2seq	10	0.09196	[0.0589, 0.125]	0.0416	5	0
tropical	seq2seq	100	0.08744	[0.062, 0.117]	0.0353	5	0
tropical	seq2seq	1000	0.1043	[0.0666, 0.146]	0.0538	5	0
tropical	seq2seq	1e+04	0.07918	[0.068, 0.0924]	0.0151	5	0
vanilla	pointer	0	0.1858	[0.13, 0.237]	0.0653	5	0
vanilla	pointer	0.01	0.1915	[0.13, 0.254]	0.0795	5	0
vanilla	pointer	0.1	0.196	[0.123, 0.262]	0.0899	5	0
vanilla	pointer	1	0.1895	[0.125, 0.254]	0.0824	5	0
vanilla	pointer	10	0.1918	[0.129, 0.253]	0.0794	5	0
vanilla	pointer	100	0.1894	[0.129, 0.245]	0.0734	5	0
vanilla	pointer	1000	0.1881	[0.12, 0.257]	0.0877	5	0
vanilla	pointer	1e+04	0.1853	[0.116, 0.26]	0.0932	5	0
vanilla	seq2seq	0	0.1539	[0.131, 0.177]	0.0307	5	0
vanilla	seq2seq	0.01	0.1851	[0.166, 0.204]	0.0245	5	0
vanilla	seq2seq	0.1	0.1845	[0.165, 0.205]	0.0265	5	0
vanilla	seq2seq	1	0.1745	[0.16, 0.189]	0.0183	5	0
vanilla	seq2seq	10	0.1799	[0.162, 0.197]	0.0224	5	0
vanilla	seq2seq	100	0.1659	[0.153, 0.18]	0.0175	5	0
vanilla	seq2seq	1000	0.1202	[0.0893, 0.151]	0.0407	5	0
vanilla	seq2seq	1e+04	0.1319	[0.115, 0.149]	0.0219	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.00782	[-0.0589, 0.0432]	0.9983
adaptive	pointer	0.1	-0.01634	[-0.0674, 0.0347]	0.9114
adaptive	pointer	1	-0.00542	[-0.0565, 0.0456]	0.9998
adaptive	pointer	10	-0.01106	[-0.0621, 0.04]	0.9870
adaptive	pointer	100	-0.01468	[-0.0657, 0.0364]	0.9450
adaptive	pointer	1000	-0.0169	[-0.0679, 0.0341]	0.8978
adaptive	pointer	1e+04	-0.02094	[-0.072, 0.0301]	0.7736
adaptive	seq2seq	0.01	-0.00412	[-0.0555, 0.0472]	1.0000
adaptive	seq2seq	0.1	-0.00948	[-0.0608, 0.0419]	0.9948
adaptive	seq2seq	1	-0.0162	[-0.0675, 0.0351]	0.9167
adaptive	seq2seq	10	0.00694	[-0.0444, 0.0583]	0.9992
adaptive	seq2seq	100	0.01566	[-0.0357, 0.067]	0.9282
adaptive	seq2seq	1000	0.01266	[-0.0387, 0.064]	0.9742
adaptive	seq2seq	1e+04	-0.01306	[-0.0644, 0.0383]	0.9698
tropical	pointer	0.01	-0.03642	[-0.152, 0.0796]	0.9186
tropical	pointer	0.1	-0.02928	[-0.145, 0.0868]	0.9710
tropical	pointer	1	-0.00956	[-0.126, 0.106]	1.0000
tropical	pointer	10	-0.00226	[-0.118, 0.114]	1.0000
tropical	pointer	100	-0.0233	[-0.139, 0.0927]	0.9915

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	pointer	1000	-0.01556	[-0.132, 0.1]	0.9993
tropical	pointer	1e+04	-0.01548	[-0.132, 0.101]	0.9993
tropical	seq2seq	0.01	-0.002	[-0.0782, 0.0742]	1.0000
tropical	seq2seq	0.1	-0.00698	[-0.0831, 0.0692]	0.9999
tropical	seq2seq	1	-0.0068	[-0.083, 0.0694]	0.9999
tropical	seq2seq	10	-0.0071	[-0.0833, 0.0691]	0.9999
tropical	seq2seq	100	-0.01162	[-0.0878, 0.0645]	0.9984
tropical	seq2seq	1000	0.00528	[-0.0709, 0.0814]	1.0000
tropical	seq2seq	1e+04	-0.01988	[-0.096, 0.0563]	0.9656
vanilla	pointer	0.01	0.0057	[-0.137, 0.149]	1.0000
vanilla	pointer	0.1	0.01018	[-0.133, 0.153]	1.0000
vanilla	pointer	1	0.0037	[-0.139, 0.147]	1.0000
vanilla	pointer	10	0.00602	[-0.137, 0.149]	1.0000
vanilla	pointer	100	0.00366	[-0.139, 0.147]	1.0000
vanilla	pointer	1000	0.00234	[-0.141, 0.145]	1.0000
vanilla	pointer	1e+04	-4.800×10^{-4}	[-0.143, 0.142]	1.0000
vanilla	seq2seq	0.01	0.03118	[-0.0147, 0.0771]	0.2944
vanilla	seq2seq	0.1	0.03054	[-0.0154, 0.0765]	0.3137
vanilla	seq2seq	1	0.02056	[-0.0254, 0.0665]	0.7033
vanilla	seq2seq	10	0.02592	[-0.02, 0.0718]	0.4784
vanilla	seq2seq	100	0.01194	[-0.034, 0.0579]	0.9662
vanilla	seq2seq	1000	-0.03378	[-0.0797, 0.0121]	0.2241
vanilla	seq2seq	1e+04	-0.02206	[-0.068, 0.0239]	0.6396

Table 15: `ood_val_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.001776	[-0.00274, -1.68×10^{-4}]	0.4302	-0.131	0.3229
adaptive	seq2seq	0.001761	[-8.20×10^{-4}, 0.00409]	0.4785	0.101	0.4587
tropical	pointer	0.002733	[-0.00406, 0.01]	0.6017	0.12	0.3749
tropical	seq2seq	-0.001212	[-0.00607, 0.00452]	0.7250	-0.0324	0.8206
vanilla	pointer	-0.001224	[-0.00277, 5.44×10^{-4}]	0.8526	-0.0533	0.6973
vanilla	seq2seq	-0.01061	[-0.0168, -0.00455]	$< 10^{-4}$	-0.493	0.0004

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.00888	[-0.0926, 0.11]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.03324	[-0.167, 0.101]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.03058	[-0.171, 0.11]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.00438	[-0.109, 0.0997]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	6.000×10^{-4}	[-0.102, 0.103]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.01808	[-0.125, 0.0887]	1.0000

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.00902	[-0.14, 0.121]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.00612	[-0.0884, 0.0762]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.05488	[-0.117, 0.00684]	0.6903	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.08806	[-0.181, 0.00457]	0.6337	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.0924	[-0.168, -0.0172]	0.3241	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.08224	[-0.147, -0.0173]	0.3191	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.0879	[-0.156, -0.0201]	0.3187	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.07844	[-0.136, -0.0211]	0.3055	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.01582	[-0.0709, 0.0393]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.0527	[-0.0923, -0.0131]	0.3148	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.07628	[0.00166, 0.151]	0.6049	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.06276	[-0.0237, 0.149]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.04976	[-0.0586, 0.158]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.06716	[-0.0285, 0.163]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.0592	[-0.0299, 0.148]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.05794	[-0.0135, 0.129]	0.8759	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.05704	[-0.0403, 0.154]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.05582	[-0.055, 0.167]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.02718	[-0.102, 0.048]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.06248	[-0.123, -0.00155]	0.6049	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.0672	[-0.116, -0.0188]	0.2922	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.06394	[-0.112, -0.0158]	0.3165	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.04616	[-0.0983, 0.00599]	0.7686	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.02346	[-0.0936, 0.0466]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.01926	[-0.0325, 0.071]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.01818	[-0.0352, -0.00117]	0.5775	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.1353	[0.0718, 0.199]	0.0735	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.1316	[0.103, 0.161]	0.0052	*

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.1284	[0.0933, 0.164]	0.0117	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.1461	[0.116, 0.176]	0.0042	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.1173	[0.0817, 0.153]	0.0166	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.105	[0.0424, 0.168]	0.1634	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.1057	[0.0595, 0.152]	0.0599	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.1274	[0.0779, 0.177]	0.0408	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.0956	[-0.0138, 0.205]	0.9395	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.06118	[-0.0531, 0.175]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.0733	[-0.048, 0.195]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.09284	[0.0187, 0.167]	0.3561	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.1004	[0.0269, 0.174]	0.2881	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.08392	[-0.0188, 0.187]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.07476	[-0.0415, 0.191]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.1	[0.0345, 0.165]	0.2120	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.03184	[-0.0144, 0.0781]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.00636	[-0.106, 0.119]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.01148	[-0.121, 0.144]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.01498	[-0.0852, 0.115]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.01194	[-0.102, 0.126]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.02356	[-0.0643, 0.111]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.06796	[-0.0323, 0.168]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.05342	[-0.0663, 0.173]	1.0000	

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.6224	[0.6, 0.645]	0.0297	5	0
adaptive	pointer	0.01	0.6344	[0.592, 0.673]	0.0521	5	0
adaptive	pointer	0.1	0.6313	[0.589, 0.669]	0.0526	5	0
adaptive	pointer	1	0.6331	[0.591, 0.666]	0.0476	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	10	0.6288	[0.586, 0.665]	0.0508	5	0
adaptive	pointer	100	0.6211	[0.568, 0.667]	0.0638	5	0
adaptive	pointer	1000	0.6229	[0.574, 0.664]	0.0595	5	0
adaptive	pointer	1e+04	0.618	[0.567, 0.661]	0.0641	5	0
adaptive	seq2seq	0	0.6269	[0.553, 0.696]	0.096	5	0
adaptive	seq2seq	0.01	0.656	[0.609, 0.693]	0.0544	5	0
adaptive	seq2seq	0.1	0.6497	[0.611, 0.689]	0.0516	5	0
adaptive	seq2seq	1	0.6562	[0.617, 0.69]	0.0466	5	0
adaptive	seq2seq	10	0.6775	[0.609, 0.721]	0.0769	5	0
adaptive	seq2seq	100	0.617	[0.534, 0.7]	0.107	5	0
adaptive	seq2seq	1000	0.5949	[0.486, 0.704]	0.137	5	0
adaptive	seq2seq	1e+04	0.5704	[0.477, 0.634]	0.104	5	0
tropical	pointer	0	0.5893	[0.56, 0.624]	0.0424	5	0
tropical	pointer	0.01	0.593	[0.553, 0.643]	0.0567	5	0
tropical	pointer	0.1	0.5769	[0.519, 0.625]	0.067	5	0
tropical	pointer	1	0.5993	[0.565, 0.647]	0.0546	5	0
tropical	pointer	10	0.584	[0.555, 0.606]	0.0335	5	0
tropical	pointer	100	0.5679	[0.512, 0.625]	0.0766	5	0
tropical	pointer	1000	0.5893	[0.567, 0.61]	0.0286	5	0
tropical	pointer	1e+04	0.5884	[0.558, 0.61]	0.0338	5	0
tropical	seq2seq	0	0.5871	[0.437, 0.691]	0.167	5	0
tropical	seq2seq	0.01	0.5447	[0.394, 0.683]	0.195	5	0
tropical	seq2seq	0.1	0.5509	[0.459, 0.663]	0.133	5	0
tropical	seq2seq	1	0.3962	[0.262, 0.546]	0.179	5	0
tropical	seq2seq	10	0.4172	[0.302, 0.527]	0.153	5	0
tropical	seq2seq	100	0.4776	[0.241, 0.676]	0.278	5	0
tropical	seq2seq	1000	0.3783	[0.157, 0.599]	0.279	5	0
tropical	seq2seq	1e+04	0.3276	[0.163, 0.52]	0.23	5	0
vanilla	pointer	0	0.5571	[0.485, 0.628]	0.0875	5	0
vanilla	pointer	0.01	0.5313	[0.435, 0.621]	0.121	5	0
vanilla	pointer	0.1	0.5205	[0.417, 0.62]	0.128	5	0
vanilla	pointer	1	0.517	[0.422, 0.612]	0.127	5	0
vanilla	pointer	10	0.5263	[0.419, 0.621]	0.127	5	0
vanilla	pointer	100	0.5236	[0.431, 0.61]	0.118	5	0
vanilla	pointer	1000	0.515	[0.425, 0.606]	0.121	5	0
vanilla	pointer	1e+04	0.5306	[0.449, 0.614]	0.107	5	0
vanilla	seq2seq	0	0.5358	[0.49, 0.579]	0.0569	5	0
vanilla	seq2seq	0.01	0.5373	[0.437, 0.631]	0.132	5	0
vanilla	seq2seq	0.1	0.5586	[0.479, 0.645]	0.11	5	0
vanilla	seq2seq	1	0.5569	[0.505, 0.609]	0.0687	5	0
vanilla	seq2seq	10	0.5664	[0.515, 0.611]	0.0617	5	0
vanilla	seq2seq	100	0.5188	[0.434, 0.614]	0.115	5	0
vanilla	seq2seq	1000	0.5336	[0.439, 0.6]	0.106	5	0
vanilla	seq2seq	1e+04	0.5801	[0.537, 0.62]	0.0549	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.01205	[-0.0815, 0.106]	0.9994
adaptive	pointer	0.1	0.008907	[-0.0846, 0.102]	0.9999
adaptive	pointer	1	0.0107	[-0.0828, 0.104]	0.9997
adaptive	pointer	10	0.006423	[-0.0871, 0.0999]	1.0000
adaptive	pointer	100	-0.001268	[-0.0948, 0.0923]	1.0000

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	1000	4.859×10^{-4}	[-0.093, 0.094]	1.0000
adaptive	pointer	1e+04	-0.00442	[-0.0979, 0.0891]	1.0000
adaptive	seq2seq	0.01	0.02907	[-0.127, 0.185]	0.9946
adaptive	seq2seq	0.1	0.02276	[-0.134, 0.179]	0.9988
adaptive	seq2seq	1	0.02927	[-0.127, 0.186]	0.9943
adaptive	seq2seq	10	0.0506	[-0.106, 0.207]	0.9070
adaptive	seq2seq	100	-0.009893	[-0.166, 0.146]	1.0000
adaptive	seq2seq	1000	-0.03199	[-0.188, 0.124]	0.9906
adaptive	seq2seq	1e+04	-0.05653	[-0.213, 0.0997]	0.8550
tropical	pointer	0.01	0.003727	[-0.0867, 0.0941]	1.0000
tropical	pointer	0.1	-0.01237	[-0.103, 0.078]	0.9992
tropical	pointer	1	0.01006	[-0.0803, 0.1]	0.9998
tropical	pointer	10	-0.00529	[-0.0957, 0.0851]	1.0000
tropical	pointer	100	-0.02136	[-0.112, 0.069]	0.9793
tropical	pointer	1000	5.105×10^{-5}	[-0.0903, 0.0904]	1.0000
tropical	pointer	1e+04	-8.351×10^{-4}	[-0.0912, 0.0896]	1.0000
tropical	seq2seq	0.01	-0.04238	[-0.406, 0.321]	0.9997
tropical	seq2seq	0.1	-0.0362	[-0.4, 0.328]	0.9999
tropical	seq2seq	1	-0.1909	[-0.555, 0.173]	0.5531
tropical	seq2seq	10	-0.1699	[-0.534, 0.194]	0.6660
tropical	seq2seq	100	-0.1095	[-0.473, 0.254]	0.9323
tropical	seq2seq	1000	-0.2088	[-0.573, 0.155]	0.4614
tropical	seq2seq	1e+04	-0.2595	[-0.623, 0.104]	0.2501
vanilla	pointer	0.01	-0.02581	[-0.232, 0.18]	0.9995
vanilla	pointer	0.1	-0.03657	[-0.242, 0.169]	0.9959
vanilla	pointer	1	-0.04004	[-0.246, 0.166]	0.9929
vanilla	pointer	10	-0.03082	[-0.237, 0.175]	0.9985
vanilla	pointer	100	-0.03344	[-0.239, 0.172]	0.9976
vanilla	pointer	1000	-0.04206	[-0.248, 0.164]	0.9906
vanilla	pointer	1e+04	-0.02648	[-0.232, 0.179]	0.9994
vanilla	seq2seq	0.01	0.001473	[-0.161, 0.164]	1.0000
vanilla	seq2seq	0.1	0.0228	[-0.139, 0.185]	0.9990
vanilla	seq2seq	1	0.02106	[-0.141, 0.183]	0.9994
vanilla	seq2seq	10	0.03056	[-0.132, 0.193]	0.9941
vanilla	seq2seq	100	-0.01698	[-0.179, 0.145]	0.9999
vanilla	seq2seq	1000	-0.002177	[-0.164, 0.16]	1.0000
vanilla	seq2seq	1e+04	0.04425	[-0.118, 0.206]	0.9573

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.002793	[-0.00538, -7.45×10^{-4}]	0.5274	-0.101	0.4545
adaptive	seq2seq	-0.01448	[-0.0374, 0.00618]	0.0502	-0.128	0.3531
tropical	pointer	-7.237×10^{-4}	[-0.00816, 0.00661]	0.8644	0.0476	0.7425
tropical	seq2seq	-0.03269	[-0.0625, -0.00101]	0.0641	-0.211	0.1134
vanilla	pointer	-2.281×10^{-4}	[-0.00453, 0.00407]	0.9809	-0.0324	0.8248
vanilla	seq2seq	0.001441	[-0.0067, 0.00915]	0.8533	0.04	0.7724

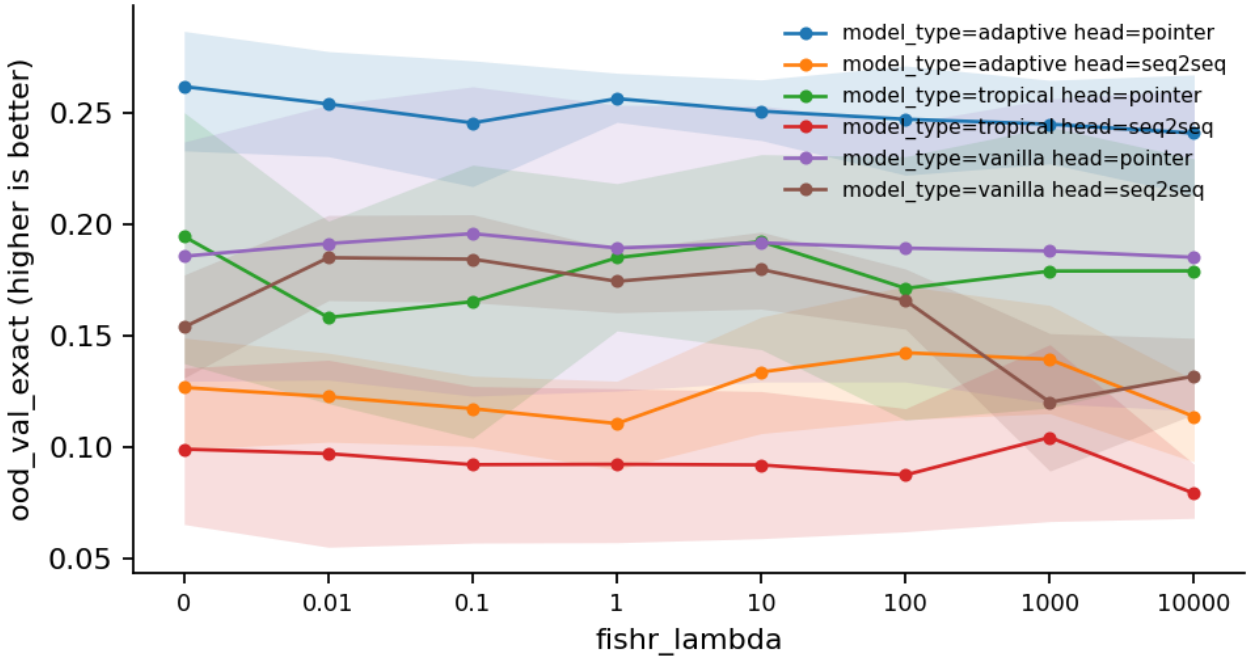


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

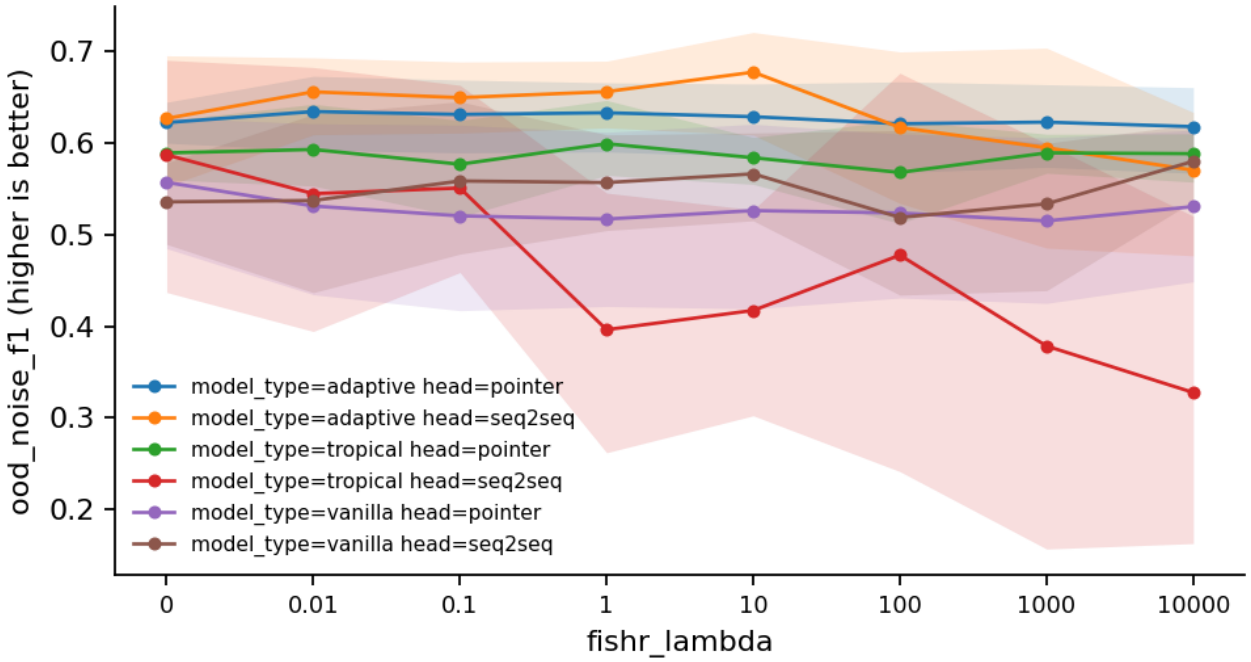


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0322	[-0.0771, 0.142]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.06174	[-0.14, 0.263]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.0564	[-0.14, 0.253]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.0823	[-0.114, 0.278]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.05773	[-0.102, 0.217]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.04428	[-0.125, 0.214]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.07432	[-0.0722, 0.221]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.05784	[-0.0898, 0.206]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.05131	[-0.0941, 0.197]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.00745	[-0.183, 0.198]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.007687	[-0.153, 0.137]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.1607	[-0.386, 0.0648]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.1491	[-0.313, 0.0143]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.04124	[-0.291, 0.209]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.1553	[-0.441, 0.131]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.2525	[-0.533, 0.0283]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.06533	[-0.0393, 0.17]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.1032	[-0.0473, 0.254]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.1108	[-0.0382, 0.26]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.1161	[-0.0394, 0.272]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.1026	[-0.0582, 0.263]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.0975	[-0.037, 0.232]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.1079	[-0.042, 0.258]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.08738	[-0.0668, 0.242]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.09108	[-0.0327, 0.215]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.1187	[-0.0687, 0.306]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.09104	[-0.057, 0.239]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.09929	[-0.0162, 0.215]	1.0000

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.1111	[0.0526, 0.17]	0.0929
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.09817	[0.0548, 0.142]	0.0526
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.06127	[-0.0584, 0.181]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.009702	[-0.0942, 0.0748]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.004495	[-0.137, 0.128]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.02152	[-0.0351, -0.0079]	0.2719
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.01835	[-0.0498, 0.0131]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.02306	[-0.0346, -0.0115]	0.1240
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.04867	[-0.157, 0.0594]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.00413	[-0.165, 0.173]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.02798	[-0.165, 0.221]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.04761	[-0.112, 0.208]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.002152	[-0.201, 0.205]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.04826	[-0.225, 0.321]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.02598	[-0.211, 0.263]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.2031	[-0.0692, 0.475]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.1667	[-0.05, 0.383]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.09032	[-0.334, 0.515]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.211	[-0.142, 0.564]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.2609	[-0.0514, 0.573]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.02126	[-0.0559, 0.0984]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.006029	[-0.161, 0.149]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.03811	[-0.209, 0.133]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.03984	[-0.234, 0.154]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.04012	[-0.222, 0.142]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.004803	[-0.136, 0.145]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.01863	[-0.156, 0.119]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.04947	[-0.18, 0.0813]	1.0000

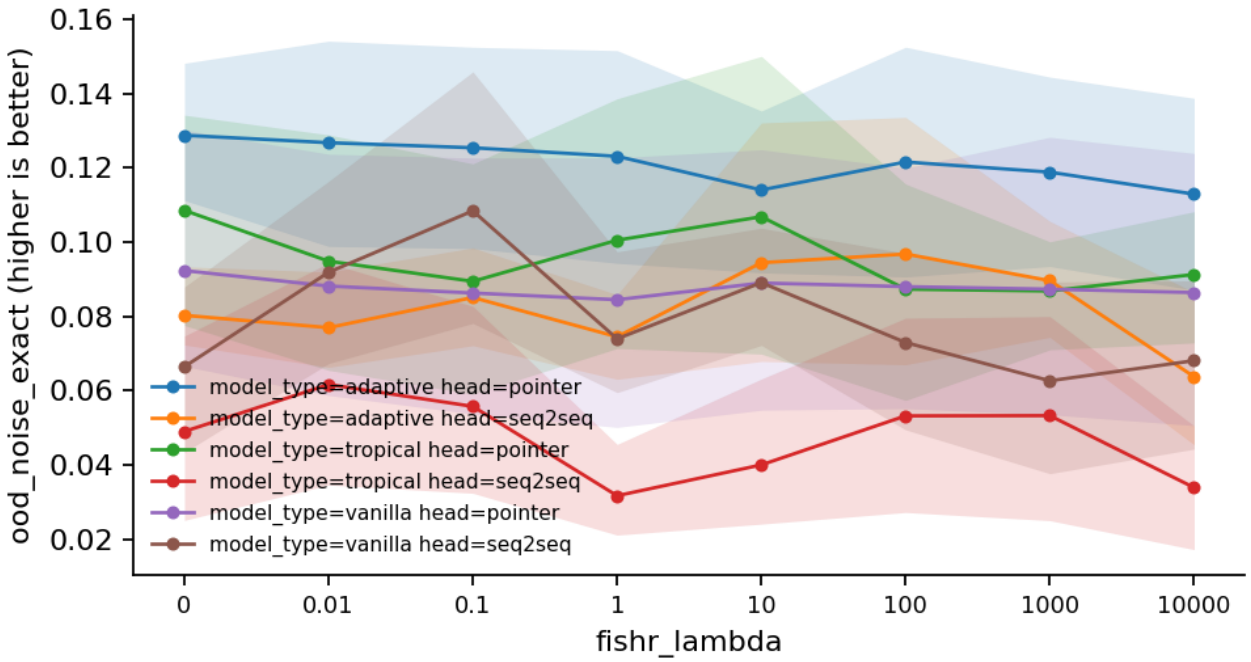


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.1287	[0.111, 0.148]	0.0243	5	0
adaptive	pointer	0.01	0.1267	[0.0988, 0.154]	0.0359	5	0
adaptive	pointer	0.1	0.1254	[0.0983, 0.152]	0.0356	5	0
adaptive	pointer	1	0.1231	[0.0943, 0.152]	0.038	5	0
adaptive	pointer	10	0.114	[0.0917, 0.135]	0.0279	5	0
adaptive	pointer	100	0.1215	[0.0906, 0.152]	0.0398	5	0
adaptive	pointer	1000	0.1188	[0.0934, 0.144]	0.033	5	0
adaptive	pointer	1e+04	0.1129	[0.0871, 0.139]	0.036	5	0
adaptive	seq2seq	0	0.08026	[0.0723, 0.0934]	0.0147	5	0
adaptive	seq2seq	0.01	0.07694	[0.0663, 0.0919]	0.0176	5	0
adaptive	seq2seq	0.1	0.08506	[0.0721, 0.0984]	0.0174	5	0
adaptive	seq2seq	1	0.07448	[0.0631, 0.0858]	0.0143	5	0
adaptive	seq2seq	10	0.09442	[0.0679, 0.132]	0.0431	5	0
adaptive	seq2seq	100	0.09676	[0.067, 0.134]	0.0448	5	0
adaptive	seq2seq	1000	0.08958	[0.0744, 0.106]	0.0204	5	0
adaptive	seq2seq	1e+04	0.06374	[0.0455, 0.0865]	0.0276	5	0
tropical	pointer	0	0.1085	[0.0776, 0.134]	0.0358	5	0
tropical	pointer	0.01	0.09484	[0.0655, 0.129]	0.0408	5	0
tropical	pointer	0.1	0.0894	[0.0591, 0.121]	0.0394	5	0
tropical	pointer	1	0.1005	[0.0714, 0.139]	0.0435	5	0
tropical	pointer	10	0.1068	[0.0699, 0.15]	0.0496	5	0
tropical	pointer	100	0.0873	[0.0574, 0.116]	0.0382	5	0
tropical	pointer	1000	0.0868	[0.071, 0.1]	0.019	5	0
tropical	pointer	1e+04	0.09126	[0.0729, 0.108]	0.0234	5	0
tropical	seq2seq	0	0.049	[0.0251, 0.0747]	0.0332	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.06164	[0.0346, 0.0944]	0.0385	5	0
tropical	seq2seq	0.1	0.05574	[0.0324, 0.0827]	0.0325	5	0
tropical	seq2seq	1	0.0317	[0.0211, 0.0455]	0.0163	5	0
tropical	seq2seq	10	0.04	[0.0241, 0.0631]	0.0254	5	0
tropical	seq2seq	100	0.0532	[0.0272, 0.0795]	0.0338	5	0
tropical	seq2seq	1000	0.05326	[0.025, 0.0799]	0.0356	5	0
tropical	seq2seq	1e+04	0.0339	[0.0172, 0.0505]	0.0217	5	0
vanilla	pointer	0	0.0923	[0.0666, 0.13]	0.0429	5	0
vanilla	pointer	0.01	0.08814	[0.0587, 0.124]	0.0412	5	0
vanilla	pointer	0.1	0.08626	[0.0537, 0.123]	0.0455	5	0
vanilla	pointer	1	0.08446	[0.0501, 0.123]	0.0469	5	0
vanilla	pointer	10	0.08896	[0.0548, 0.125]	0.0463	5	0
vanilla	pointer	100	0.08798	[0.0552, 0.12]	0.0426	5	0
vanilla	pointer	1000	0.0873	[0.0536, 0.128]	0.0475	5	0
vanilla	pointer	1e+04	0.08638	[0.0506, 0.124]	0.0481	5	0
vanilla	seq2seq	0	0.06642	[0.0434, 0.0879]	0.029	5	0
vanilla	seq2seq	0.01	0.09184	[0.0673, 0.116]	0.0328	5	0
vanilla	seq2seq	0.1	0.1084	[0.0781, 0.146]	0.0463	5	0
vanilla	seq2seq	1	0.07394	[0.0595, 0.0974]	0.0259	5	0
vanilla	seq2seq	10	0.08906	[0.0722, 0.104]	0.0211	5	0
vanilla	seq2seq	100	0.07294	[0.0495, 0.097]	0.0307	5	0
vanilla	seq2seq	1000	0.06264	[0.0377, 0.0893]	0.0326	5	0
vanilla	seq2seq	1e+04	0.06806	[0.0443, 0.0896]	0.0297	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.002	[-0.0617, 0.0577]	1.0000
adaptive	pointer	0.1	-0.00336	[-0.063, 0.0563]	1.0000
adaptive	pointer	1	-0.00566	[-0.0653, 0.054]	0.9999
adaptive	pointer	10	-0.01472	[-0.0744, 0.045]	0.9742
adaptive	pointer	100	-0.00718	[-0.0669, 0.0525]	0.9996
adaptive	pointer	1000	-0.00992	[-0.0696, 0.0498]	0.9972
adaptive	pointer	1e+04	-0.01584	[-0.0755, 0.0438]	0.9626
adaptive	seq2seq	0.01	-0.00332	[-0.0514, 0.0448]	1.0000
adaptive	seq2seq	0.1	0.0048	[-0.0433, 0.0529]	0.9999
adaptive	seq2seq	1	-0.00578	[-0.0539, 0.0423]	0.9996
adaptive	seq2seq	10	0.01416	[-0.034, 0.0623]	0.9390
adaptive	seq2seq	100	0.0165	[-0.0316, 0.0646]	0.8822
adaptive	seq2seq	1000	0.00932	[-0.0388, 0.0574]	0.9931
adaptive	seq2seq	1e+04	-0.01652	[-0.0646, 0.0316]	0.8818
tropical	pointer	0.01	-0.01366	[-0.0791, 0.0518]	0.9895
tropical	pointer	0.1	-0.0191	[-0.0845, 0.0463]	0.9412
tropical	pointer	1	-0.00802	[-0.0734, 0.0574]	0.9996
tropical	pointer	10	-0.0017	[-0.0671, 0.0637]	1.0000
tropical	pointer	100	-0.0212	[-0.0866, 0.0442]	0.9067
tropical	pointer	1000	-0.0217	[-0.0871, 0.0437]	0.8972
tropical	pointer	1e+04	-0.01724	[-0.0827, 0.0482]	0.9640
tropical	seq2seq	0.01	0.01264	[-0.0406, 0.0659]	0.9788
tropical	seq2seq	0.1	0.00674	[-0.0465, 0.06]	0.9995
tropical	seq2seq	1	-0.0173	[-0.0705, 0.0359]	0.9056
tropical	seq2seq	10	-0.009	[-0.0622, 0.0442]	0.9969
tropical	seq2seq	100	0.0042	[-0.049, 0.0574]	1.0000

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	seq2seq	1000	0.00426	[-0.049, 0.0575]	1.0000
tropical	seq2seq	1e+04	-0.0151	[-0.0683, 0.0381]	0.9486
vanilla	pointer	0.01	-0.00416	[-0.0831, 0.0748]	1.0000
vanilla	pointer	0.1	-0.00604	[-0.085, 0.0729]	1.0000
vanilla	pointer	1	-0.00784	[-0.0868, 0.0711]	0.9999
vanilla	pointer	10	-0.00334	[-0.0823, 0.0756]	1.0000
vanilla	pointer	100	-0.00432	[-0.0833, 0.0746]	1.0000
vanilla	pointer	1000	-0.005	[-0.0839, 0.0739]	1.0000
vanilla	pointer	1e+04	-0.00592	[-0.0849, 0.073]	1.0000
vanilla	seq2seq	0.01	0.02542	[-0.03, 0.0809]	0.6827
vanilla	seq2seq	0.1	0.04196	[-0.0135, 0.0974]	0.2015
vanilla	seq2seq	1	0.00752	[-0.0479, 0.063]	0.9992
vanilla	seq2seq	10	0.02264	[-0.0328, 0.0781]	0.7772
vanilla	seq2seq	100	0.00652	[-0.0489, 0.062]	0.9997
vanilla	seq2seq	1000	-0.00378	[-0.0592, 0.0517]	1.0000
vanilla	seq2seq	1e+04	0.00164	[-0.0538, 0.0571]	1.0000

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.002006	[-0.00292, -0.00111]	0.4732	-0.0895	0.5039
adaptive	seq2seq	-2.957×10^{-4}	[-0.00491, 0.00432]	0.9045	-0.0171	0.9048
tropical	pointer	-0.00104	[-0.00539, 0.00244]	0.7301	-0.0019	1.0000
tropical	seq2seq	-0.002381	[-0.00984, 0.00244]	0.3457	-0.177	0.1886
vanilla	pointer	1.143×10^{-5}	[-0.002, 0.00249]	0.9974	-0.0533	0.6931
vanilla	seq2seq	-0.005851	[-0.0103, -0.00146]	0.0328	-0.23	0.0864

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0162	[-0.0355, 0.0679]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.0067	[-0.0763, 0.0897]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.00314	[-0.0672, 0.0735]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.01602	[-0.0602, 0.0922]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.01784	[-0.0695, 0.105]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-6.800×10^{-4}	[-0.0597, 0.0583]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-5.000×10^{-4}	[-0.048, 0.047]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.00488	[-0.0504, 0.0602]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.01742	[-0.0729, 0.038]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.0302	[-0.0783, 0.0179]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.05264	[-0.129, 0.0242]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.04224	[-0.0788, -0.00566]	0.4579
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.04906	[-0.0901, -0.00802]	0.4520
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.01974	[-0.036, -0.00344]	0.4520
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.00938	[-0.0689, 0.0501]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.03416	[-0.0887, 0.0204]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.03642	[-0.0201, 0.0929]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.03858	[-0.0282, 0.105]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.0391	[-0.0268, 0.105]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.0386	[-0.0317, 0.109]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.02504	[-0.0474, 0.0975]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.03356	[-0.0235, 0.0906]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.0315	[-0.0377, 0.101]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.0265	[-0.0504, 0.103]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.01384	[-0.0311, 0.0588]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.0149	[-0.0401, 0.0103]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.02332	[-0.0714, 0.0248]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	5.400×10^{-4}	[-0.0278, 0.0289]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.00536	[-0.0405, 0.0512]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.02382	[-0.0375, 0.0852]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.02694	[-0.012, 0.0659]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.00432	[-0.0278, 0.0192]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.04846	[0.0141, 0.0828]	0.3807
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.04978	[0.016, 0.0835]	0.3427
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.0403	[-0.00354, 0.0841]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.04858	[0.0061, 0.0911]	0.6996
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.01958	[-0.0583, 0.0975]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.02478	[-0.0607, 0.11]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.02922	[-0.013, 0.0715]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.04914	[0.00383, 0.0945]	0.7378
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.0595	[0.00496, 0.114]	0.7378
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.0332	[-0.0593, 0.126]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.03366	[-0.0533, 0.121]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.06878	[0.00886, 0.129]	0.6996
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.0668	[-0.0114, 0.145]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.0341	[-0.037, 0.105]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.03354	[-0.00504, 0.0721]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.05736	[0.0233, 0.0915]	0.2286
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.02588	[-0.0158, 0.0675]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.0037	[-0.0719, 0.0645]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.02212	[-0.104, 0.0602]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.01052	[-0.0658, 0.0868]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-1.000×10^{-4}	[-0.0761, 0.0759]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.01504	[-0.0305, 0.0606]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.02466	[-0.0495, 0.0988]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.01832	[-0.0534, 0.0901]	1.0000

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8466	[0.844, 0.849]	0.00361	5	0
adaptive	pointer	0.01	0.851	[0.846, 0.856]	0.00681	5	0
adaptive	pointer	0.1	0.8511	[0.846, 0.857]	0.00702	5	0
adaptive	pointer	1	0.8512	[0.846, 0.856]	0.00651	5	0
adaptive	pointer	10	0.8511	[0.846, 0.856]	0.00657	5	0
adaptive	pointer	100	0.8516	[0.846, 0.857]	0.00692	5	0
adaptive	pointer	1000	0.8525	[0.847, 0.858]	0.00769	5	0
adaptive	pointer	1e+04	0.8538	[0.847, 0.861]	0.00893	5	0
adaptive	seq2seq	0	0.7834	[0.781, 0.786]	0.00322	5	0
adaptive	seq2seq	0.01	0.7826	[0.781, 0.785]	0.00253	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0.1	0.7828	[0.78, 0.785]	0.00305	5	0
adaptive	seq2seq	1	0.7827	[0.78, 0.785]	0.003	5	0
adaptive	seq2seq	10	0.783	[0.781, 0.785]	0.00265	5	0
adaptive	seq2seq	100	0.7842	[0.781, 0.789]	0.00479	5	0
adaptive	seq2seq	1000	0.7817	[0.78, 0.784]	0.00275	5	0
adaptive	seq2seq	1e+04	0.7741	[0.771, 0.777]	0.00425	5	0
tropical	pointer	0	0.9274	[0.911, 0.943]	0.0204	5	0
tropical	pointer	0.01	0.9218	[0.909, 0.934]	0.0166	5	0
tropical	pointer	0.1	0.9247	[0.911, 0.936]	0.0163	5	0
tropical	pointer	1	0.922	[0.901, 0.938]	0.0248	5	0
tropical	pointer	10	0.921	[0.909, 0.931]	0.0137	5	0
tropical	pointer	100	0.919	[0.9, 0.934]	0.023	5	0
tropical	pointer	1000	0.9189	[0.903, 0.932]	0.0187	5	0
tropical	pointer	1e+04	0.9187	[0.902, 0.931]	0.0192	5	0
tropical	seq2seq	0	0.8435	[0.832, 0.856]	0.0152	5	0
tropical	seq2seq	0.01	0.8555	[0.846, 0.863]	0.0104	5	0
tropical	seq2seq	0.1	0.8562	[0.845, 0.864]	0.0116	5	0
tropical	seq2seq	1	0.8529	[0.845, 0.861]	0.0109	5	0
tropical	seq2seq	10	0.8513	[0.841, 0.86]	0.0118	5	0
tropical	seq2seq	100	0.8492	[0.839, 0.858]	0.0122	5	0
tropical	seq2seq	1000	0.8313	[0.803, 0.852]	0.0318	5	0
tropical	seq2seq	1e+04	0.8016	[0.761, 0.83]	0.0449	5	0
vanilla	pointer	0	0.8522	[0.848, 0.856]	0.00537	5	0
vanilla	pointer	0.01	0.8552	[0.852, 0.858]	0.00346	5	0
vanilla	pointer	0.1	0.8556	[0.853, 0.858]	0.00305	5	0
vanilla	pointer	1	0.8554	[0.853, 0.858]	0.00305	5	0
vanilla	pointer	10	0.8553	[0.853, 0.858]	0.00328	5	0
vanilla	pointer	100	0.8554	[0.853, 0.858]	0.0033	5	0
vanilla	pointer	1000	0.8558	[0.853, 0.859]	0.00344	5	0
vanilla	pointer	1e+04	0.8581	[0.855, 0.861]	0.00414	5	0
vanilla	seq2seq	0	0.7931	[0.781, 0.804]	0.0141	5	0
vanilla	seq2seq	0.01	0.7936	[0.781, 0.806]	0.0156	5	0
vanilla	seq2seq	0.1	0.7932	[0.782, 0.804]	0.0145	5	0
vanilla	seq2seq	1	0.7925	[0.781, 0.804]	0.014	5	0
vanilla	seq2seq	10	0.7928	[0.783, 0.803]	0.0139	5	0
vanilla	seq2seq	100	0.7896	[0.781, 0.802]	0.014	5	0
vanilla	seq2seq	1000	0.7866	[0.78, 0.797]	0.0117	5	0
vanilla	seq2seq	1e+04	0.7762	[0.772, 0.783]	0.00792	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.004422	[-0.00763, 0.0165]	0.8472
adaptive	pointer	0.1	0.004536	[-0.00752, 0.0166]	0.8323
adaptive	pointer	1	0.004648	[-0.00741, 0.0167]	0.8167
adaptive	pointer	10	0.004526	[-0.00753, 0.0166]	0.8335
adaptive	pointer	100	0.005063	[-0.00699, 0.0171]	0.7558
adaptive	pointer	1000	0.005969	[-0.00609, 0.018]	0.6108
adaptive	pointer	1e+04	0.007245	[-0.00481, 0.0193]	0.4142
adaptive	seq2seq	0.01	-8.623×10^{-4}	[-0.00674, 0.00502]	0.9987
adaptive	seq2seq	0.1	-5.981×10^{-4}	[-0.00648, 0.00528]	0.9999
adaptive	seq2seq	1	-7.256×10^{-4}	[-0.00661, 0.00516]	0.9996
adaptive	seq2seq	10	-4.405×10^{-4}	[-0.00632, 0.00544]	1.0000

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	seq2seq	100	7.306×10^{-4}	[-0.00515, 0.00661]	0.9996	
adaptive	seq2seq	1000	-0.0017	[-0.00758, 0.00418]	0.9438	
adaptive	seq2seq	1e+04	-0.009373	[-0.0153, -0.00349]	0.0007	*
tropical	pointer	0.01	-0.005681	[-0.0395, 0.0282]	0.9970	
tropical	pointer	0.1	-0.002717	[-0.0366, 0.0311]	1.0000	
tropical	pointer	1	-0.005464	[-0.0393, 0.0284]	0.9977	
tropical	pointer	10	-0.006443	[-0.0403, 0.0274]	0.9938	
tropical	pointer	100	-0.00846	[-0.0423, 0.0254]	0.9723	
tropical	pointer	1000	-0.008507	[-0.0424, 0.0253]	0.9715	
tropical	pointer	1e+04	-0.008788	[-0.0426, 0.0251]	0.9665	
tropical	seq2seq	0.01	0.01198	[-0.0266, 0.0506]	0.9224	
tropical	seq2seq	0.1	0.01267	[-0.0259, 0.0513]	0.9014	
tropical	seq2seq	1	0.009382	[-0.0292, 0.048]	0.9760	
tropical	seq2seq	10	0.00781	[-0.0308, 0.0464]	0.9912	
tropical	seq2seq	100	0.00563	[-0.033, 0.0442]	0.9988	
tropical	seq2seq	1000	-0.01222	[-0.0508, 0.0264]	0.9155	
tropical	seq2seq	1e+04	-0.04196	[-0.0805, -0.00337]	0.0286	*
vanilla	pointer	0.01	0.003017	[-0.00346, 0.00949]	0.6684	
vanilla	pointer	0.1	0.003464	[-0.00301, 0.00994]	0.5339	
vanilla	pointer	1	0.003222	[-0.00325, 0.0097]	0.6062	
vanilla	pointer	10	0.003136	[-0.00334, 0.00961]	0.6321	
vanilla	pointer	100	0.003223	[-0.00325, 0.0097]	0.6060	
vanilla	pointer	1000	0.003644	[-0.00283, 0.0101]	0.4820	
vanilla	pointer	1e+04	0.005903	$[-5.74 \times 10^{-4}$, 0.0124]	0.0859	
vanilla	seq2seq	0.01	5.151×10^{-4}	[-0.0229, 0.0239]	1.0000	
vanilla	seq2seq	0.1	1.334×10^{-4}	[-0.0233, 0.0236]	1.0000	
vanilla	seq2seq	1	-5.367×10^{-4}	[-0.024, 0.0229]	1.0000	
vanilla	seq2seq	10	-2.149×10^{-4}	[-0.0236, 0.0232]	1.0000	
vanilla	seq2seq	100	-0.003464	[-0.0269, 0.02]	0.9986	
vanilla	seq2seq	1000	-0.006436	[-0.0299, 0.017]	0.9557	
vanilla	seq2seq	1e+04	-0.01688	[-0.0403, 0.00654]	0.2413	

Table 27: `id_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	4.197×10^{-4}	$[1.06 \times 10^{-4}, 6.70 \times 10^{-4}]$	0.4637	0.116	0.3827
adaptive	seq2seq	-9.386×10^{-4}	$[-0.00125, -6.29 \times 10^{-4}]$	0.0099	-0.288	0.0282
tropical	pointer	-8.533×10^{-4}	$[-0.00174, 4.13 \times 10^{-5}]$	0.5734	-0.0819	0.5553
tropical	seq2seq	-0.007691	[-0.0124, -0.00433]	0.0005	-0.482	0.0006
vanilla	pointer	3.221×10^{-4}	$[1.79 \times 10^{-4}, 4.66 \times 10^{-4}]$	0.2442	0.192	0.1488
vanilla	seq2seq	-0.002438	$[-0.00404, -9.38 \times 10^{-4}]$	0.0295	-0.28	0.0398

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.07527	[0.0475, 0.103]	0.0155	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.06658	[0.0451, 0.0881]	0.0131	*

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.06909	[0.0487, 0.0894]	0.0106	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.06659	[0.0363, 0.0969]	0.0155	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.06569	[0.0488, 0.0826]	0.0067	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.06359	[0.0363, 0.0908]	0.0155	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.06312	[0.0403, 0.086]	0.0155	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.06058	[0.0387, 0.0825]	0.0155	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.05046	[0.0239, 0.0771]	0.0187	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.06193	[0.0419, 0.082]	0.0131	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.063	[0.0443, 0.0817]	0.0106	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.06038	[0.0356, 0.0852]	0.0155	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.05849	[0.036, 0.081]	0.0155	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.05956	[0.0387, 0.0804]	0.0151	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.04468	[0.00764, 0.0817]	0.0572	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.02539	[-0.0282, 0.0789]	0.2584	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.005623	[-0.0108, -4.59×10^{-4}]	0.6248	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.004218	[-0.011, 0.00255]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.004551	[-0.0111, 0.00196]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.004196	[-0.0105, 0.00213]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.004233	[-0.0102, 0.00178]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.003782	[-0.0114, 0.0038]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.003297	[-0.0109, 0.00427]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.004281	[-0.0145, 0.00592]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.009627	[-0.0277, 0.00845]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.011	[-0.0306, 0.0086]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01036	[-0.0287, 0.00797]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.009816	[-0.028, 0.00839]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.009853	[-0.0283, 0.0086]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.005432	[-0.0243, 0.0135]	1.0000	

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.00489	[-0.0203, 0.0105]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.00212	[-0.0149, 0.0107]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.06312	[0.0562, 0.07]	0.0003	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.06841	[0.0591, 0.0777]	0.0007	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.06826	[0.0591, 0.0774]	0.0007	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.0685	[0.0593, 0.0777]	0.0007	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.06809	[0.0582, 0.078]	0.0008	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.06745	[0.0545, 0.0804]	0.0022	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.07079	[0.0591, 0.0825]	0.0013	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.07974	[0.0689, 0.0906]	0.0007	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.08393	[0.0593, 0.109]	0.0090	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.06627	[0.0422, 0.0904]	0.0091	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.06854	[0.0459, 0.0912]	0.0091	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.06908	[0.0454, 0.0928]	0.0091	*
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.06968	[0.0501, 0.0892]	0.0082	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.06984	[0.0398, 0.0999]	0.0091	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.08764	[0.0455, 0.13]	0.0091	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.1171	[0.0582, 0.176]	0.0091	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.05912	[0.0409, 0.0773]	0.0091	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.06162	[0.04, 0.0832]	0.0091	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.06245	[0.0428, 0.0821]	0.0091	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.06288	[0.044, 0.0817]	0.0091	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.06247	[0.0433, 0.0816]	0.0091	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.0658	[0.0453, 0.0863]	0.0091	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.0692	[0.0516, 0.0868]	0.0060	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.0819	[0.0683, 0.0955]	0.0013	*

A.7 `id_exact`

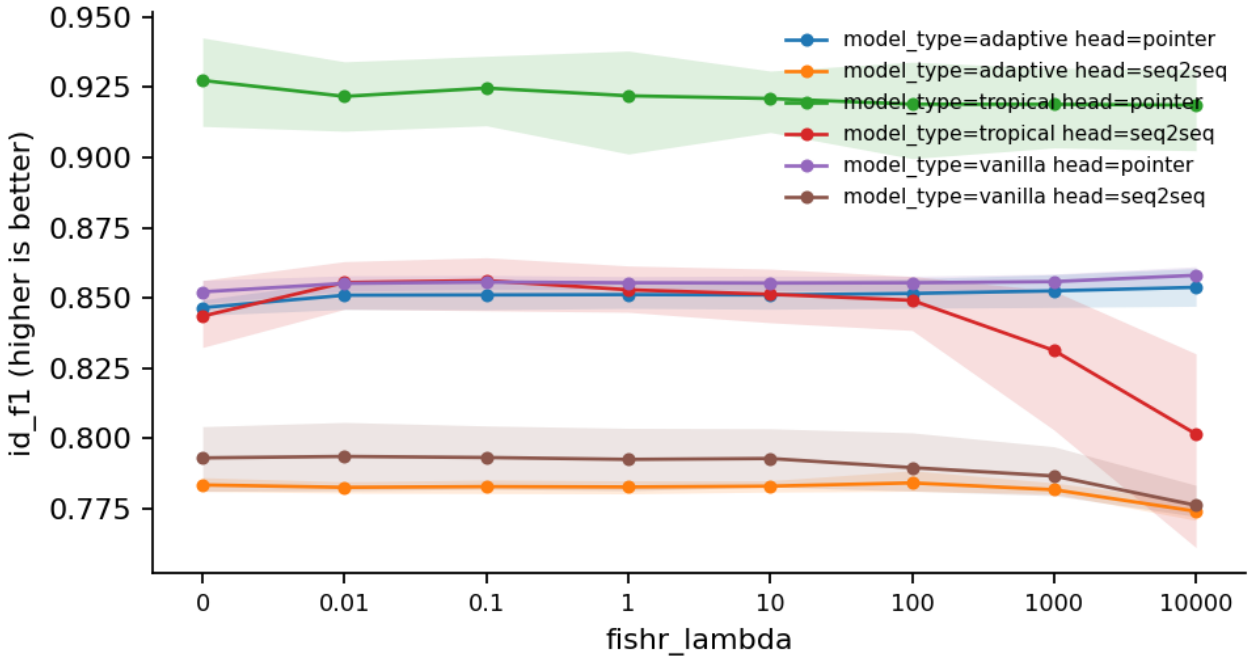


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

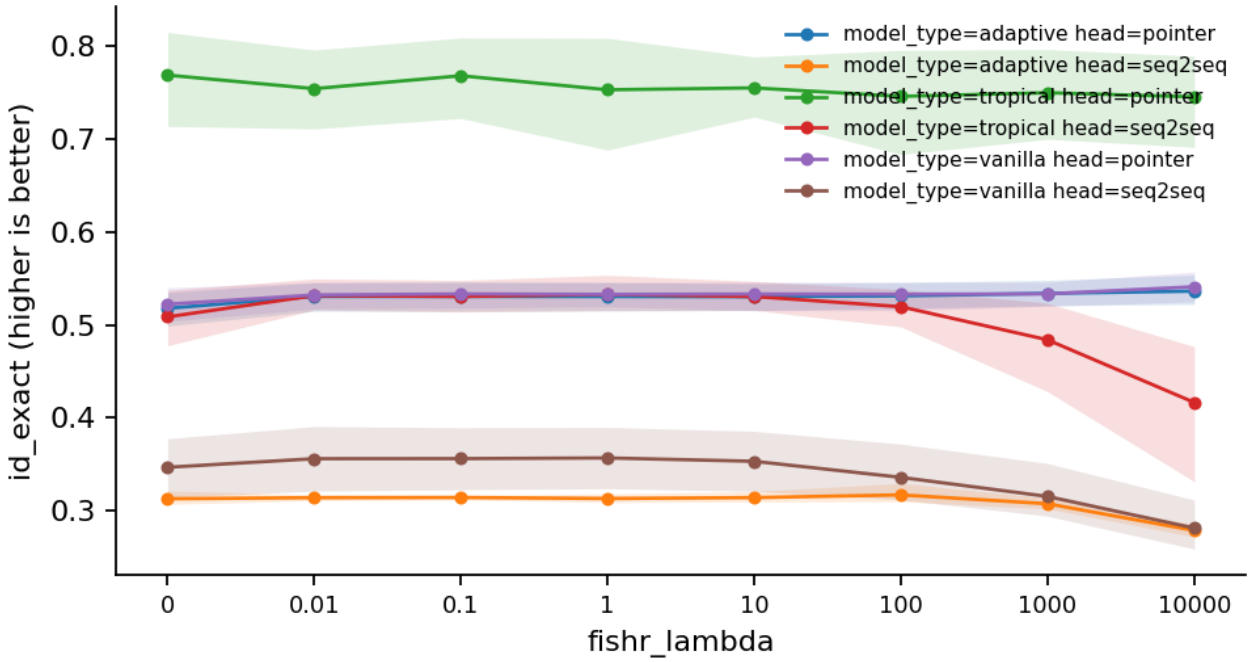


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.5179	[0.499, 0.535]	0.0237	5	0
adaptive	pointer	0.01	0.5308	[0.515, 0.545]	0.0194	5	0
adaptive	pointer	0.1	0.5307	[0.515, 0.547]	0.021	5	0
adaptive	pointer	1	0.5307	[0.515, 0.546]	0.0216	5	0
adaptive	pointer	10	0.5305	[0.516, 0.545]	0.0199	5	0
adaptive	pointer	100	0.5313	[0.518, 0.546]	0.0188	5	0
adaptive	pointer	1000	0.534	[0.521, 0.548]	0.0181	5	0
adaptive	pointer	1e+04	0.5364	[0.522, 0.554]	0.0202	5	0
adaptive	seq2seq	0	0.3131	[0.307, 0.322]	0.00972	5	0
adaptive	seq2seq	0.01	0.3143	[0.311, 0.318]	0.00433	5	0
adaptive	seq2seq	0.1	0.3145	[0.312, 0.317]	0.00341	5	0
adaptive	seq2seq	1	0.3134	[0.309, 0.318]	0.00573	5	0
adaptive	seq2seq	10	0.3143	[0.31, 0.321]	0.00762	5	0
adaptive	seq2seq	100	0.3173	[0.31, 0.33]	0.0136	5	0
adaptive	seq2seq	1000	0.3078	[0.302, 0.314]	0.00727	5	0
adaptive	seq2seq	1e+04	0.279	[0.272, 0.286]	0.00851	5	0
tropical	pointer	0	0.7687	[0.714, 0.815]	0.0663	5	0
tropical	pointer	0.01	0.7542	[0.711, 0.796]	0.058	5	0
tropical	pointer	0.1	0.7681	[0.722, 0.809]	0.0558	5	0
tropical	pointer	1	0.753	[0.688, 0.809]	0.0808	5	0
tropical	pointer	10	0.755	[0.724, 0.788]	0.0415	5	0
tropical	pointer	100	0.7459	[0.683, 0.796]	0.0753	5	0
tropical	pointer	1000	0.7498	[0.7, 0.796]	0.0625	5	0
tropical	pointer	1e+04	0.7454	[0.691, 0.789]	0.0645	5	0
tropical	seq2seq	0	0.5087	[0.478, 0.537]	0.0404	5	0
tropical	seq2seq	0.01	0.5316	[0.516, 0.55]	0.0222	5	0
tropical	seq2seq	0.1	0.5308	[0.514, 0.548]	0.0217	5	0
tropical	seq2seq	1	0.5329	[0.516, 0.554]	0.0247	5	0
tropical	seq2seq	10	0.5306	[0.516, 0.547]	0.0201	5	0
tropical	seq2seq	100	0.5197	[0.498, 0.538]	0.0253	5	0
tropical	seq2seq	1000	0.484	[0.428, 0.523]	0.0625	5	0
tropical	seq2seq	1e+04	0.4167	[0.332, 0.477]	0.0953	5	0
vanilla	pointer	0	0.5223	[0.504, 0.54]	0.022	5	0
vanilla	pointer	0.01	0.5323	[0.517, 0.546]	0.0187	5	0
vanilla	pointer	0.1	0.5334	[0.518, 0.546]	0.0171	5	0
vanilla	pointer	1	0.5329	[0.52, 0.545]	0.0176	5	0
vanilla	pointer	10	0.5333	[0.516, 0.546]	0.0185	5	0
vanilla	pointer	100	0.5331	[0.516, 0.546]	0.0182	5	0
vanilla	pointer	1000	0.5335	[0.52, 0.546]	0.0175	5	0
vanilla	pointer	1e+04	0.5411	[0.525, 0.557]	0.0219	5	0
vanilla	seq2seq	0	0.3469	[0.315, 0.378]	0.0393	5	0
vanilla	seq2seq	0.01	0.3562	[0.321, 0.391]	0.0455	5	0
vanilla	seq2seq	0.1	0.3563	[0.323, 0.39]	0.0446	5	0
vanilla	seq2seq	1	0.357	[0.324, 0.39]	0.0425	5	0
vanilla	seq2seq	10	0.3534	[0.321, 0.386]	0.0426	5	0
vanilla	seq2seq	100	0.3361	[0.312, 0.372]	0.0396	5	0
vanilla	seq2seq	1000	0.3156	[0.295, 0.351]	0.04	5	0
vanilla	seq2seq	1e+04	0.2816	[0.259, 0.312]	0.0354	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	0.01286	[-0.0228, 0.0485]	0.8559	
adaptive	pointer	0.1	0.0128	[-0.0228, 0.0484]	0.8586	
adaptive	pointer	1	0.01274	[-0.0229, 0.0484]	0.8610	
adaptive	pointer	10	0.01256	[-0.0231, 0.0482]	0.8684	
adaptive	pointer	100	0.01342	[-0.0222, 0.049]	0.8312	
adaptive	pointer	1000	0.0161	[-0.0195, 0.0517]	0.6950	
adaptive	pointer	1e+04	0.01848	[-0.0171, 0.0541]	0.5645	
adaptive	seq2seq	0.01	0.00116	[-0.013, 0.0153]	1.0000	
adaptive	seq2seq	0.1	0.00136	[-0.0128, 0.0155]	0.9999	
adaptive	seq2seq	1	2.800×10^{-4}	[-0.0139, 0.0145]	1.0000	
adaptive	seq2seq	10	0.00124	[-0.0129, 0.0154]	1.0000	
adaptive	seq2seq	100	0.00416	[-0.01, 0.0183]	0.9397	
adaptive	seq2seq	1000	-0.00534	[-0.0195, 0.00883]	0.8313	
adaptive	seq2seq	1e+04	-0.0341	[-0.0483, -0.0199]	$< 10^{-4}$	*
tropical	pointer	0.01	-0.01454	[-0.126, 0.0974]	0.9994	
tropical	pointer	0.1	-6.600×10^{-4}	[-0.113, 0.111]	1.0000	
tropical	pointer	1	-0.01576	[-0.128, 0.0962]	0.9990	
tropical	pointer	10	-0.01372	[-0.126, 0.0982]	0.9996	
tropical	pointer	100	-0.02282	[-0.135, 0.0891]	0.9908	
tropical	pointer	1000	-0.01892	[-0.131, 0.093]	0.9969	
tropical	pointer	1e+04	-0.02336	[-0.135, 0.0886]	0.9895	
tropical	seq2seq	0.01	0.02288	[-0.0582, 0.104]	0.9498	
tropical	seq2seq	0.1	0.02204	[-0.059, 0.103]	0.9580	
tropical	seq2seq	1	0.02422	[-0.0569, 0.105]	0.9346	
tropical	seq2seq	10	0.0219	[-0.0592, 0.103]	0.9593	
tropical	seq2seq	100	0.01094	[-0.0701, 0.092]	0.9992	
tropical	seq2seq	1000	-0.02468	[-0.106, 0.0564]	0.9289	
tropical	seq2seq	1e+04	-0.09202	[-0.173, -0.0109]	0.0208	*
vanilla	pointer	0.01	0.00996	[-0.0233, 0.0432]	0.9337	
vanilla	pointer	0.1	0.01106	[-0.0222, 0.0443]	0.8958	
vanilla	pointer	1	0.0106	[-0.0226, 0.0438]	0.9128	
vanilla	pointer	10	0.01096	[-0.0223, 0.0442]	0.8996	
vanilla	pointer	100	0.01074	[-0.0225, 0.044]	0.9078	
vanilla	pointer	1000	0.01114	[-0.0221, 0.0444]	0.8926	
vanilla	pointer	1e+04	0.01878	[-0.0145, 0.052]	0.4776	
vanilla	seq2seq	0.01	0.00926	[-0.0629, 0.0814]	0.9994	
vanilla	seq2seq	0.1	0.00932	[-0.0628, 0.0814]	0.9994	
vanilla	seq2seq	1	0.01006	[-0.0621, 0.0822]	0.9991	
vanilla	seq2seq	10	0.00646	[-0.0657, 0.0786]	0.9999	
vanilla	seq2seq	100	-0.01082	[-0.0829, 0.0613]	0.9985	
vanilla	seq2seq	1000	-0.03136	[-0.103, 0.0408]	0.7281	
vanilla	seq2seq	1e+04	-0.06534	[-0.137, 0.00678]	0.0887	

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	8.621×10^{-4}	$[2.71 \times 10^{-5}, 0.00161]$	0.5821	0.0495	0.7237
adaptive	seq2seq	-0.004118	[-0.00508, -0.00328]	0.0003	-0.491	0.0002
tropical	pointer	-0.002501	$[-0.00487, -1.34 \times 10^{-4}]$	0.6191	-0.0438	0.7580
tropical	seq2seq	-0.01612	[-0.0271, -0.0082]	0.0006	-0.45	0.0012
vanilla	pointer	9.557×10^{-4}	$[5.24 \times 10^{-4}, 0.00138]$	0.5164	0.137	0.3109

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
vanilla	seq2seq	-0.01164	[-0.0182, -0.00509]	0.0018	-0.413	0.0028

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.2464	[0.168, 0.324]	0.0094	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.2219	[0.157, 0.287]	0.0094	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.2347	[0.176, 0.294]	0.0057	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.22	[0.134, 0.306]	0.0133	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.2217	[0.178, 0.265]	0.0024	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.2128	[0.134, 0.291]	0.0133	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.2163	[0.152, 0.281]	0.0094	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.2043	[0.145, 0.264]	0.0094	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.1618	[0.1, 0.223]	0.0133	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.1754	[0.104, 0.247]	0.0133	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.1745	[0.109, 0.24]	0.0133	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.1759	[0.111, 0.241]	0.0133	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.1772	[0.116, 0.238]	0.0115	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.1835	[0.127, 0.24]	0.0094	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.1685	[0.0826, 0.254]	0.0133	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.1351	[0.0225, 0.248]	0.0291	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.00442	[-0.0262, 0.0174]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.00152	[-0.0152, 0.0122]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.00268	[-0.0168, 0.0115]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.00228	[-0.0155, 0.0109]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.00282	[-0.0131, 0.00742]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.00174	[-0.0136, 0.0102]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	5.400×10^{-4}	[-0.012, 0.0131]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.00472	[-0.0226, 0.0132]	1.0000	

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.03384	[-0.0885, 0.0208]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.04194	[-0.0998, 0.016]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.0418	[-0.0979, 0.0143]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.04362	[-0.0977, 0.0105]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.03906	[-0.0941, 0.016]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.01886	[-0.0764, 0.0387]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.00782	[-0.0602, 0.0446]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.0026	[-0.0491, 0.0439]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.2048	[0.177, 0.232]	0.0005	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.2165	[0.192, 0.241]	0.0003	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.2163	[0.19, 0.242]	0.0004	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.2173	[0.194, 0.241]	0.0003	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.2161	[0.192, 0.24]	0.0003	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.2141	[0.188, 0.24]	0.0004	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.2263	[0.2, 0.253]	0.0004	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.2574	[0.236, 0.279]	0.0001	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.26	[0.177, 0.343]	0.0104	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.2226	[0.164, 0.281]	0.0060	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.2373	[0.181, 0.293]	0.0045	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.22	[0.128, 0.312]	0.0122	*
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.2244	[0.178, 0.27]	0.0028	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.2263	[0.137, 0.315]	0.0122	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.2658	[0.182, 0.35]	0.0104	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.3287	[0.188, 0.47]	0.0122	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.1754	[0.12, 0.231]	0.0104	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.1761	[0.105, 0.247]	0.0122	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.1771	[0.107, 0.247]	0.0122	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.1759	[0.11, 0.242]	0.0122	*

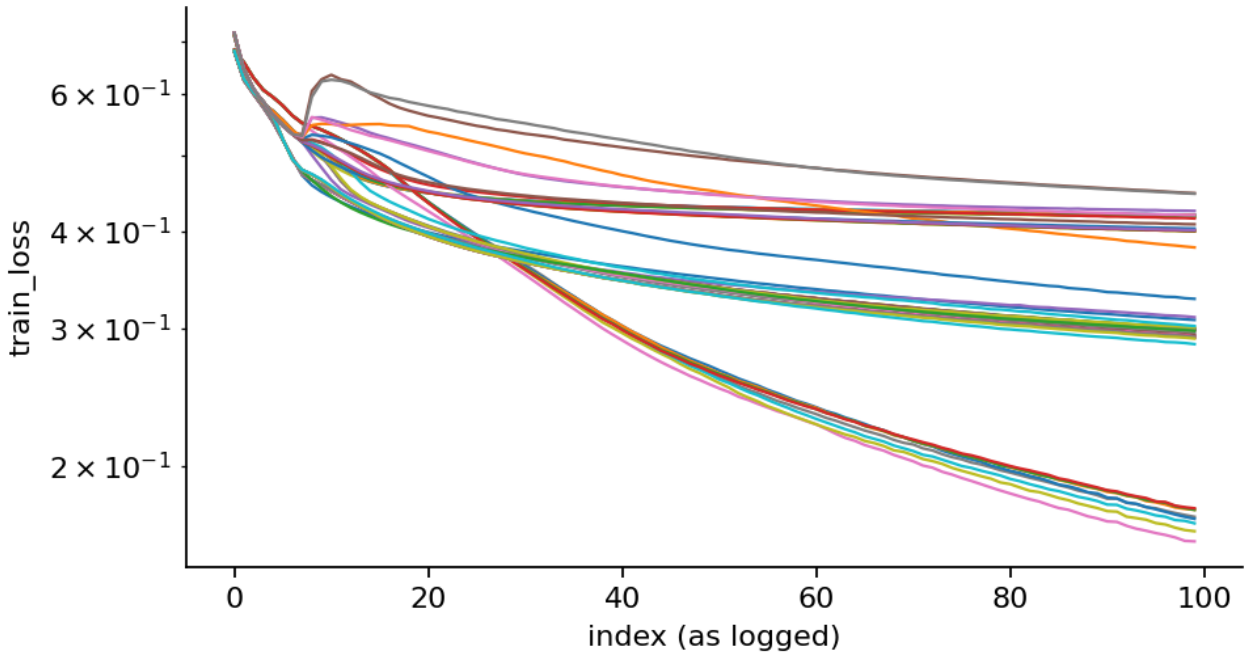


Figure 9: `train_loss`, the mean over seeds per cell.

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.1799	[0.113, 0.247]	0.0122	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.197	[0.13, 0.264]	0.0104	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.2179	[0.15, 0.286]	0.0104	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.2595	[0.192, 0.327]	0.0060	*

B Curves

C Configuration

The base configuration every run shares (frozen hash `69813b292d06`); factor levels are in Table 1.

```
task:
  name: LISDataset
  value_range:
  - 1
  - 10
  target_range: null
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
data:
  train_lengths:
```

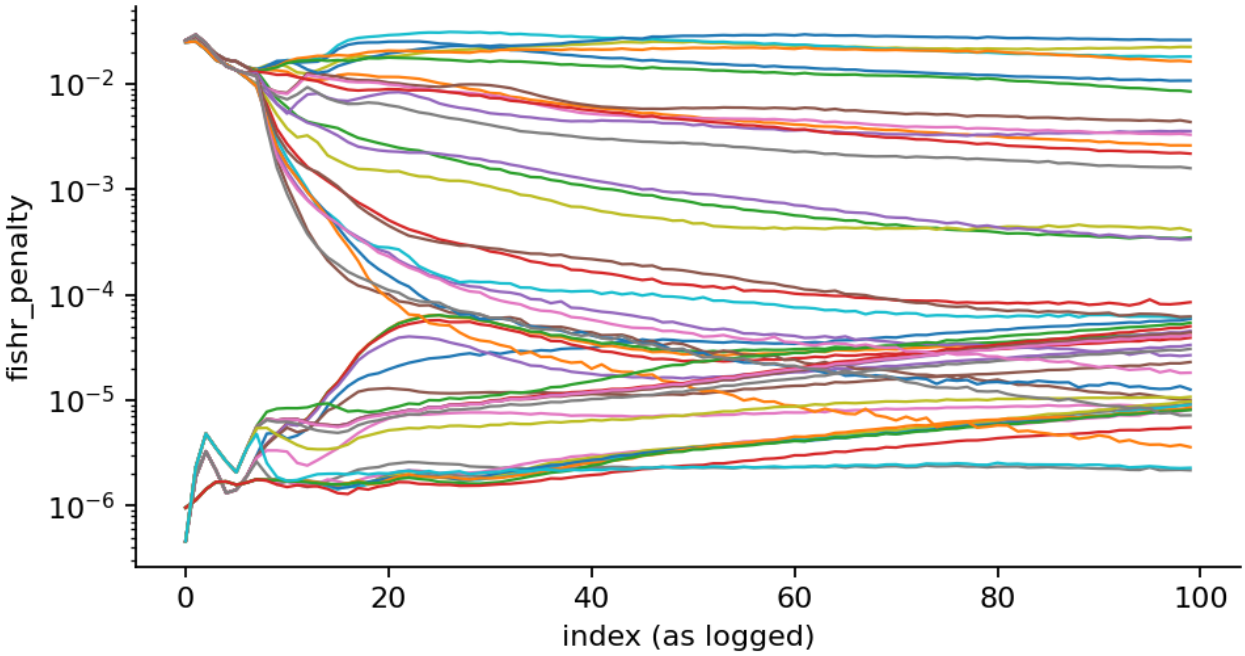


Figure 10: `fishr_penalty`, the mean over seeds per cell.

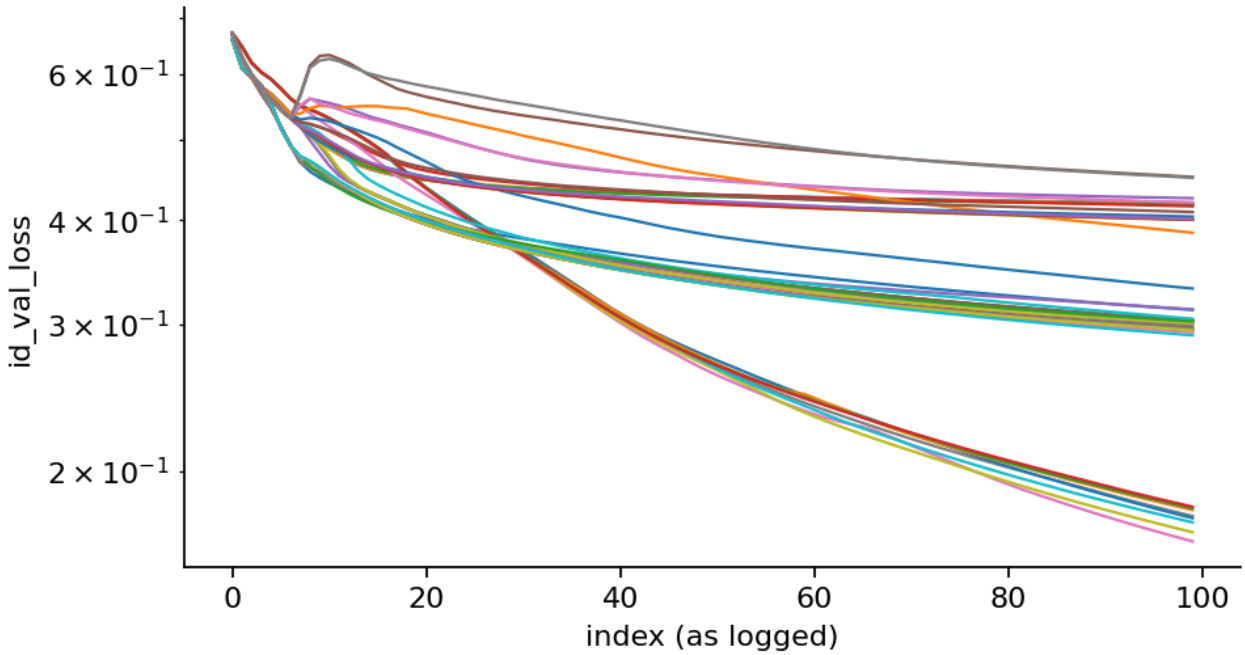


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range: null
  ood_noise_range: null
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/LIS/results`.